

Chapter 7

Observability and state observers

The concepts of observability and reconstructability refer to the possibility of determining the state of a system by making use of the associated input and output data. These concepts prove to be particularly important in those control problems where it is required to know the state of the system at every time instant. The state is not always available or measurable but, as we will see, observability and reconstructability are strictly related to the possibility of realising algorithms that can “estimate” the state value in real time.

Estimating the state at some time t can be regarded as an *observability* problem when the available input and output data correspond to time instants subsequent to t , while in the *reconstructability* problems the estimate is obtained by using data corresponding to time instants preceding t .

7.1 Observability of linear, discrete-time systems

Definition 7.1.1 [NON-OBSERVABLE STATES IN $[0, k-1]$] *Given an n -dimensional linear, discrete-time system $\Sigma = (F, G, H)$, a state $\mathbf{x} \in \mathbb{R}^n$ is not observable in $[0, k-1]$ if for every input sequence*

$$\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-2),$$

the output sequences $\mathbf{y}(\cdot)$ and $\mathbf{y}'(\cdot)$ obtained corresponding to that input sequence and to the initial states $\mathbf{x}$ and $\mathbf{0}$, respectively, coincide at times $0, 1, \dots, k-1$, namely if

$$\begin{aligned} \mathbf{y}(t) &= \sum_{\sigma=0}^{t-1} HF^{t-1-\sigma} G \mathbf{u}(\sigma) + HF^t \mathbf{x} \\ &= \sum_{\sigma=0}^{t-1} HF^{t-1-\sigma} G \mathbf{u}(\sigma) = \mathbf{y}'(t), \quad t = 0, 1, \dots, k-1. \end{aligned} \quad (7.1)$$

Therefore, $\mathbf{x}$ is not observable in $[0, k-1]$ if no “input/output experiment” can be conceived in the interval $[0, k-1]$ allowing to determine whether the initial state of the system is $\mathbf{x}$ or $\mathbf{0}$.

It is easily seen that, independently of the input sequence, condition (11.6) is equivalent to

$$0 = \mathbf{y}(t) - \mathbf{y}'(t) = HF^t \mathbf{x}, \quad t = 0, 1, \dots, k-1, \quad (7.2)$$

namely

$$\mathbf{x} \in \ker \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{k-1} \end{bmatrix} =: \ker \mathcal{O}_{[0,k-1]}. \quad (7.3)$$

$\mathcal{O}_{[0,k-1]} \in \mathbb{R}^{kp \times n}$ is called *observability matrix corresponding to the interval* $[0, k-1]$ and it associates the state $\mathbf{x} \in X$ with the vector

$$\mathcal{O}_{[0,k-1]}\mathbf{x} = \begin{bmatrix} H\mathbf{x} \\ HF\mathbf{x} \\ \vdots \\ HF^{k-1}\mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{y}_\ell(0) \\ \mathbf{y}_\ell(1) \\ \vdots \\ \mathbf{y}_\ell(k-1) \end{bmatrix},$$

that represents the unforced output sequence generated by the initial condition $\mathbf{x}$ in the interval $[0, k-1]$. From the previous considerations, it follows that

- to determine whether a state $\mathbf{x}$ is not observable in $[0, k-1]$ amounts to verify whether it induces in the interval $[0, k-1]$ the zero (unforced) output evolution, that clearly coincides with the one induced by the state $\mathbf{0}$;
- the set of states that are not observable in $[0, k-1]$, being the kernel of the matrix $\mathcal{O}_{[0,k-1]}$, is a vector subspace of X , and we will refer to it as *non-observable subspace* in $[0, k-1]$ of the system and denote it by $X_{[0,k-1]}^{\text{no}}$:

$$X_{[0,k-1]}^{\text{no}} = \ker \mathcal{O}_{[0,k-1]}. \quad (7.4)$$

System Σ is said to be *observable* in $[0, k-1]$ if the subspace $X_{[0,k-1]}^{\text{no}}$ includes only the vector $\mathbf{0}$ or, equivalently, if the matrix $\mathcal{O}_{[0,k-1]}$ has rank n .

The non-observable subspace in $[0, k-1]$ includes the non-observable subspace in $[0, k]$: this follows from the structure of $\mathcal{O}_{[0,k-1]}$ and $\mathcal{O}_{[0,k]}$ or, simply, from the fact that if a state induces the zero unforced output evolution till time k , then it induces the zero unforced output evolution till time $k-1$. We thus get the chain of inclusions

$$X_{[0,0]}^{\text{no}} \supseteq X_{[0,1]}^{\text{no}} \supseteq \dots \supseteq X_{[0,k-1]}^{\text{no}} \supseteq X_{[0,k]}^{\text{no}} \supseteq \dots \quad (7.5)$$

Lemma 7.1.2 *If $X_{[0,k-1]}^{\text{no}} = X_{[0,k]}^{\text{no}}$, then $X_{[0,k]}^{\text{no}} = X_{[0,k+1]}^{\text{no}}$.*

PROOF Keeping into account (7.5), it is sufficient to verify that if $X_{[0,k-1]}^{\text{no}} = X_{[0,k]}^{\text{no}}$, then every $\mathbf{x} \in X_{[0,k]}^{\text{no}}$ belongs also to $X_{[0,k+1]}^{\text{no}}$. In fact, if $\mathbf{x}$ belongs to $X_{[0,k]}^{\text{no}}$, the state $F\mathbf{x}$ belongs to $X_{[0,k-1]}^{\text{no}}$ and hence, by assumption, to $X_{[0,k]}^{\text{no}}$. Therefore $\mathbf{x}$ belongs to $X_{[0,k+1]}^{\text{no}}$, because it induces the zero unforced evolution at the time instants $0, 1, \dots, k+1$. ■

Proposition 7.1.3 [NON-OBSERVABLE SUBSPACE] *In an n -dimensional system $\Sigma = (F, G, H)$, the subspace chain (7.5) becomes stationary from the n -th step onward, i.e.,*

$$X_{[0,0]}^{\text{no}} \supseteq X_{[0,1]}^{\text{no}} \supseteq \dots \supseteq X_{[0,n-1]}^{\text{no}} = X_{[0,n]}^{\text{no}} = \dots \quad (7.6)$$

The subspace

$$X^{\text{no}} := X_{[0,n-1]}^{\text{no}} = X_{[0,n]}^{\text{no}} = \dots \quad (7.7)$$

is called *non-observable subspace* of the system and it consists of the initial states that produce the zero unforced output evolution over the (discrete) half-line $[0, +\infty)$.

PROOF If H is the zero matrix, the chain (7.5) becomes stationary since the very first step.

If $H \neq 0$, the subspace $X_{[0,0]}^{\text{no}} = \ker H$ has dimension not greater than $n - 1$. If all the inclusions in (7.5) are strict till the subspace $X_{[0,n-1]}^{\text{no}}$, this latter is the null space and the chain (7.5) obviously becomes stationary after the n -th step; if, on the contrary, there is an equality in the chain before the n -th step, it is sufficient to apply Lemma 7.1.2. ■

The non-observable subspace X^{no} of the system $\Sigma = (F, G, H)$ (or, for short, of the pair (F, H)) is therefore the kernel of the observability matrix in n steps, having dimension $np \times n$. It will be referred to as *observability matrix* and denoted by $\mathcal{O}$

$$\mathcal{O} := \mathcal{O}_{[0,n-1]} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix}. \quad (7.8)$$

The subspace X^{no} includes only the zero vector $\mathbf{0}$ if and only if the observability matrix has rank n : if this is the case, the system is called *observable* and the zero state is the only one to induce an identically zero unforced output evolution. The smallest integer ω corresponding to which the matrix $\mathcal{O}_{[0,\omega-1]}$ has full rank n is called *observability index*.

Example 7.1.1 Consider the discrete-time system, with a single output, described in Figure 7.1.1.

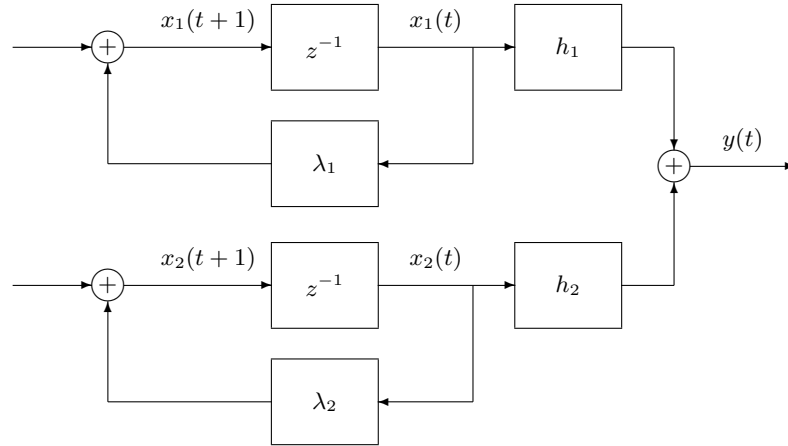


Figure 7.1.1

From the unforced state equations

$$\begin{aligned} x_1(t+1) &= \lambda_1 x_1(t), \\ x_2(t+1) &= \lambda_2 x_2(t), \\ y(t) &= h_1 x_1(t) + h_2 x_2(t), \end{aligned}$$

one can derive the matrices F and H

$$F = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \quad H = [h_1 \quad h_2],$$

and hence the observability matrix

$$\mathcal{O} = \begin{bmatrix} H \\ HF \end{bmatrix} = \begin{bmatrix} h_1 & h_2 \\ \lambda_1 h_1 & \lambda_2 h_2 \end{bmatrix}.$$

$\mathcal{O}$ has rank 2, and hence the system is observable, if and only if h_1 and h_2 are both nonzero and $\lambda_1 \neq \lambda_2$.

It is easy to see that if, for instance, $h_1 \neq 0$ and $h_2 = 0$ then $X^{\text{no}} = \text{span} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, while if h_1 and h_2 are both nonzero but $\lambda_1 = \lambda_2 = \lambda$, then the non-observable space is

$$X^{\text{no}} = \ker \mathcal{O} = \ker \begin{bmatrix} h_1 & h_2 \\ \lambda h_1 & \lambda h_2 \end{bmatrix} = \left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : h_1 x_1 + h_2 x_2 = 0 \right\}.$$

If the system is observable in $[0, k-1]$, it is possible to determine the initial state by making use of the input sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-2)$ and the output sequence $\mathbf{y}(0), \mathbf{y}(1), \dots, \mathbf{y}(k-1)$. We note, first of all, that the unforced output is given by

$$\mathbf{y}_\ell(\cdot) = \mathbf{y}(\cdot) - \mathbf{y}_f(\cdot)$$

and that the forced output $\mathbf{y}_f(\cdot)$ can be evaluated based on the input $\mathbf{u}(\cdot)$. Therefore we can determine $\mathbf{y}_\ell(0), \mathbf{y}_\ell(1), \dots, \mathbf{y}_\ell(k-1)$ from the available data and reduce ourselves to the problem of determining the initial state from $\mathbf{y}_\ell(\cdot)$.

As the vector that represents the unforced output in $[0, k-1]$ belongs to the image of $\mathcal{O}_{[0, k-1]}$, the equation

$$\begin{bmatrix} \mathbf{y}_\ell(0) \\ \mathbf{y}_\ell(1) \\ \vdots \\ \mathbf{y}_\ell(k-1) \end{bmatrix} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{k-1} \end{bmatrix} \mathbf{x}(0) = \mathcal{O}_{[0, k-1]} \mathbf{x}(0) \quad (7.9)$$

can be solved for some initial state $\mathbf{x}(0)$. Moreover, as the kernel of $\mathcal{O}_{[0, k-1]}$ is the null subspace, such initial state is unique, i.e., it is the only solution of (7.9).

Premultiplying, then, both sides of (7.9) by $\mathcal{O}_{[0, k-1]}^T$ and keeping into account that the square matrix $\mathcal{O}_{[0, k-1]}^T \mathcal{O}_{[0, k-1]}$ is invertible, since its kernel coincides with the kernel of $\mathcal{O}_{[0, k-1]}$, it follows that the initial state can be determined through the relationship

$$\mathbf{x}(0) = [\mathcal{O}_{[0, k-1]}^T \mathcal{O}_{[0, k-1]}]^{-1} \mathcal{O}_{[0, k-1]}^T \begin{bmatrix} \mathbf{y}_\ell(0) \\ \mathbf{y}_\ell(1) \\ \vdots \\ \mathbf{y}_\ell(k-1) \end{bmatrix} = \mathcal{V}_k^{-1} \mathcal{O}_{[0, k-1]}^T \begin{bmatrix} \mathbf{y}_\ell(0) \\ \mathbf{y}_\ell(1) \\ \vdots \\ \mathbf{y}_\ell(k-1) \end{bmatrix}. \quad (7.10)$$

The matrix $\mathcal{V}_k := \mathcal{O}_{[0, k-1]}^T \mathcal{O}_{[0, k-1]}$ is called the *observability gramian in $[0, k-1]$* .

Remark 1 If the system is observable in $[0, k-1]$, but the output data $\tilde{\mathbf{y}}_\ell(\cdot)$ are noisy, the values introduced in equation (7.9) to evaluate $\mathbf{x}(0)$ are not those corresponding to the unforced output evolution generated by the initial state, and may not even belong to $\text{Im} \mathcal{O}_{[0, k-1]}$. In this latter case, they do not represent any admissible unforced output sequence and equation (7.9) has no solutions. The state one deduces from (7.10) corresponding to noisy data however has some relevance in the problem solution, since it is the vector in $\mathbb{R}^n$ that generates the unforced output trajectory which is closest in norm to the measured noisy output $\tilde{\mathbf{y}}_\ell(\cdot)$.

Indeed, suppose that our aim is to determine the initial state $\mathbf{x}(0)$ that generates the unforced output evolution that “differs less” from the available noisy output

$$\tilde{\mathbf{Y}}_\ell := \begin{bmatrix} \tilde{\mathbf{y}}_\ell(0) \\ \tilde{\mathbf{y}}_\ell(1) \\ \vdots \\ \tilde{\mathbf{y}}_\ell(k-1) \end{bmatrix}.$$

This amounts to solving with respect to $\mathbf{x}(0)$ the following minimum problem

$$\min_{\mathbf{x}(0)} \|\tilde{\mathbf{Y}}_\ell - \mathcal{O}_{[0,k-1]}\mathbf{x}(0)\|_2, \quad (7.11)$$

namely, upon decomposing $\tilde{\mathbf{Y}}_\ell$ in the form $\tilde{\mathbf{Y}}_\ell = \mathbf{Y}_1 + \mathbf{Y}_2$, with $\mathbf{Y}_1 \in \text{Im}\mathcal{O}_{[0,k-1]}$ and $\mathbf{Y}_2 \in \text{Im}\mathcal{O}_{[0,k-1]}^\perp = \ker \mathcal{O}_{[0,k-1]}^T$, to solve the equivalent problem

$$\min_{\mathbf{x}(0)} \|\mathbf{Y}_1 - \mathcal{O}_{[0,k-1]}\mathbf{x}(0)\|_2^2 + \|\mathbf{Y}_2\|_2^2. \quad (7.12)$$

It is obvious that the minimum value of (7.12) is obtained corresponding to the only state $\mathbf{x}(0)$ that solves equation

$$\mathcal{O}_{[0,k-1]}\mathbf{x}(0) = \mathbf{Y}_1. \quad (7.13)$$

Since $\mathcal{O}_{[0,k-1]}^T \mathbf{Y}_2 = \mathbf{0}$, the state $\mathbf{x}(0)$ that solves (7.13) also satisfies the equation

$$\mathcal{O}_{[0,k-1]}^T \mathcal{O}_{[0,k-1]}\mathbf{x}(0) = \mathcal{O}_{[0,k-1]}^T \mathbf{Y}_1 = \mathcal{O}_{[0,k-1]}^T (\mathbf{Y}_1 + \mathbf{Y}_2) = \mathcal{O}_{[0,k-1]}^T \tilde{\mathbf{Y}}_\ell. \quad (7.14)$$

As the gramian $\mathcal{V}_k = \mathcal{O}_{[0,k-1]}^T \mathcal{O}_{[0,k-1]}$ is a non-singular matrix, equation (7.14) has the vector $\mathcal{V}_k^{-1} \mathcal{O}_{[0,k-1]}^T \tilde{\mathbf{Y}}_\ell$ as its unique solution. Therefore $\mathcal{V}_k^{-1} \mathcal{O}_{[0,k-1]}^T \tilde{\mathbf{Y}}_\ell$ is the solution of (7.13).

- EXERCISE 7.1.1 Suppose that system $\Sigma = (F, G, H)$ is not observable in $[0, k-1]$.
 - i) If $\mathbf{y}_\ell(\cdot)$ is a (not noisy) unforced output, equations $\mathcal{O}_{[0,k-1]}\mathbf{x} = \tilde{\mathbf{Y}}_\ell$ and $\mathcal{O}_{[0,k-1]}^T \mathcal{O}_{[0,k-1]}\mathbf{x} = \mathcal{O}_{[0,k-1]}^T \tilde{\mathbf{Y}}_\ell$ have the same (not empty) set of solutions, which is an *indistinguishability class* in $[0, k-1]$, specifically the one consisting of all the initial states that induce the unforced output sequence $\mathbf{y}_\ell(t)$, $t = 0, 1, \dots, k-1$.
 - ii) If $\tilde{\mathbf{y}}_\ell(\cdot)$ is a noisy unforced output, the first equation may not have solutions, while the second one has always solutions. Overall, the set of states that solve the equations represent an indistinguishability class in $[0, k-1]$, and the unforced evolution induced by any of them is the best approximation in norm of the noisy unforced output $\tilde{\mathbf{y}}_\ell(\cdot)$.

‡ *Hint.* (i) The solutions of (7.13) are also solutions of (7.14). If $\mathbf{x}$ solves the second equation, then $\mathcal{O}_{[0,k-1]}^T (\mathcal{O}_{[0,k-1]}\mathbf{x} - \tilde{\mathbf{Y}}_\ell) = \mathbf{0}$, and therefore $\mathcal{O}_{[0,k-1]}\mathbf{x} - \tilde{\mathbf{Y}}_\ell$, belonging to $\text{Im}\mathcal{O}_{[0,k-1]}$ and to $\ker \mathcal{O}_{[0,k-1]}^T = (\text{Im}\mathcal{O}_{[0,k-1]})^\perp$, is zero. (ii) The second member of equation (7.14) belongs to $\text{Im}\mathcal{O}_{[0,k-1]}^T = \text{Im}(\mathcal{O}_{[0,k-1]}^T \mathcal{O}_{[0,k-1]})$, and hence a solution exists. It can be identified up to a vector in $\ker(\mathcal{O}_{[0,k-1]}^T \mathcal{O}_{[0,k-1]}) = \ker \mathcal{O}_{[0,k-1]} = X_{[0,k-1]}^{\text{no}}$. To achieve optimality, one can proceed as in the case of a system which is observable in $[0, k-1]$.

Remark 2 If $\Sigma = (F, G, H)$ is observable in $[0, k-1]$, let $N = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]$ denote an orthonormal matrix whose columns represent an orthonormal basis of eigenvectors

for the positive definite matrix $\mathcal{V}_k$. By proceeding as in section 5.4 for the reachability gramian, we can express $\mathcal{V}_k$ as

$$\mathcal{V}_k = N \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\} N^T, \quad (7.15)$$

where the singular values τ_i are the square roots of the (positive) eigenvalues of $\mathcal{V}_k$, that we may assume sorted by decreasing order.

By assuming

$$\mathbf{Y}_\ell := \begin{bmatrix} \mathbf{y}_\ell(0) \\ \mathbf{y}_\ell(1) \\ \vdots \\ \mathbf{y}_\ell(k-1) \end{bmatrix},$$

from (7.9) we obtain

$$\|\mathbf{Y}_\ell\|_2^2 = \mathbf{Y}_\ell^T \mathbf{Y}_\ell = \mathbf{x}(0)^T \mathcal{V}_k \mathbf{x}(0) = \mathbf{x}(0)^T N \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\} N^T \mathbf{x}(0) \quad (7.16)$$

and if we regard $\|\mathbf{Y}_\ell\|_2^2$ as the *energy of the unforced output* $\mathcal{O}_{[0,k-1]} \mathbf{x}(0)$, induced by the state $\mathbf{x}(0)$, it is clear that τ_i^2 represents the unforced output energy corresponding to the initial state $\mathbf{x}(0) = \mathbf{v}_i$.

From the inequalities $\tau_1 \geq \tau_2 \geq \dots \geq \tau_n$, it follows that $\mathbf{v}_1$ and $\mathbf{v}_n$ represent the states with unitary norms that induce in the time interval $[0, k-1]$ the maximum energy and minimum energy unforced evolutions, respectively.

In the algebraically equivalent system $\hat{\Sigma} = (\hat{F}, \hat{G}, \hat{H}) := (N^T F N, N^T G, H N)$ the observability matrix in $[0, k-1]$ is $\hat{\mathcal{O}}_{[0,k-1]} = \mathcal{O}_{[0,k-1]} N$ and the corresponding gramian matrix is diagonal:

$$\hat{\mathcal{V}}_k = (N^T \mathcal{O}_k^T) (\mathcal{O}_{[0,k-1]} N) = N^T [N \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\} N^T] N = \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\}. \quad (7.17)$$

Since the state $\hat{\mathbf{x}}(0) = N^T \mathbf{x}(0) = N^{-1} \mathbf{x}(0)$ is associated with an unforced output having energy

$$\|\mathbf{y}_\ell\|_2^2 = \hat{\mathbf{x}}(0)^T \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\} \hat{\mathbf{x}}(0),$$

the states $\hat{\mathbf{x}}(0) = \mathbf{e}_1$ and $\hat{\mathbf{x}}(0) = \mathbf{e}_n$ of system $\hat{\Sigma}$ induce unforced outputs in $[0, k-1]$ of maximum energy τ_1^2 and minimum energy τ_n^2 , respectively.

By referring again to $\hat{\Sigma}$, the initial states that induce in $[0, k-1]$ unforced output trajectories $\mathbf{y}_\ell$ with unitary energy belong to the ellipsoidal hyper-surface

$$E^O = \{\hat{\mathbf{x}} : \hat{\mathbf{x}}^T \text{diag}\{\tau_1^2, \tau_2^2, \dots, \tau_n^2\} \hat{\mathbf{x}} = 1\}.$$

- EXERCISE 7.1.2 [NORMALISATION WITH RESPECT TO THE OUTPUT] By referring to system $\Sigma = (F, G, H)$ in Remark 2, set $S := \text{diag}\{\tau_1, \tau_2, \dots, \tau_n\}$ and $T := S N$. Verify that system

$$\Sigma^{OUT} = (F^{OUT}, G^{OUT}, H^{OUT}) := (T^{-1} F T, T^{-1} G, H T),$$

that is algebraically equivalent to Σ and that we will call “normalised with respect to the output”, has observability matrix in $[0, k-1]$ $\mathcal{O}_{[0,k-1]}^{OUT} = S^{-1} M^T \mathcal{O}_{[0,k-1]}$ and its gramian matrix is the identity, i.e., $\mathcal{V}_k^{OUT} = I_n$. In the state-space model Σ^{OUT} all the states with unitary norm induce an unforced output evolution whose first k samples have unitary norm.

If the system Σ is observable and internally stable, and $\mathbf{Y}_\ell$ denotes the sequence $\{\mathbf{y}_\ell(t)\}_{t \in \mathbb{N}}$, the energy of the unforced output over the infinite horizon:

$$\|\mathbf{Y}_\ell\|_2^2 = \sum_{t=0}^{\infty} \mathbf{y}_\ell^T(t) \mathbf{y}_\ell(t) = \sum_{t=0}^{\infty} \mathbf{x}(0)^T F^t H^T H F^t \mathbf{x}(0)$$

is a finite quantity, independently of the initial state. It is expressed by the quadratic form

$$\|\mathbf{Y}_\ell\|_2^2 = \mathbf{x}_0^T \mathcal{V}_\infty \mathbf{x}_0, \quad (7.18)$$

where the “observability gramian over infinite horizon” $\mathcal{V}_\infty$ is the unique positive definite solution of the Lyapunov equation

$$F^T X F - X = -H^T H. \quad (7.19)$$

The result follows directly from Exercise 7.1.3 (analogous to Lemma 5.5.1).

- EXERCISE 7.1.3 [LYAPUNOV EQUATION AND OBSERVABILITY] Given a discrete-time system $\Sigma = (F, -, H)$, any pair of the following statements implies the remaining one:
 - i) Σ is observable;
 - ii) the Lyapunov equation $F^T X F - X = -H^T H$ admits a positive definite solution;
 - iii) $|\Lambda(F)| < 1$, namely all the eigenvalues of F have modulus smaller than 1.

The notion of non-observability refers, as we have seen, to the impossibility of establishing, based on the input/output behaviour, whether the system initial state coincides with the zero state $\mathbf{0}$. It represents a special case of the indistinguishability notion¹ between two arbitrary states, that can be studied by making use of the results so far obtained.

Definition 7.1.4 [INDISTINGUISHABILITY] *The states $\mathbf{x}'$ and $\mathbf{x}''$ are indistinguishable in $[0, k-1]$ (notation: $\mathbf{x}' \underset{[0, k-1]}{\sim} \mathbf{x}''$) if for every input sequence $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-2)$ the associated output sequences $\mathbf{y}'(\cdot)$ and $\mathbf{y}''(\cdot)$ that correspond to the initial state $\mathbf{x}'$ and to the initial state $\mathbf{x}''$, respectively, coincide at the time instants $0, 1, \dots, k-1$, namely if*

$$\begin{aligned} \mathbf{y}'(t) &= H F^t \mathbf{x}' + \sum_{\sigma=0}^{t-1} H F^{t-1-\sigma} G \mathbf{u}(\sigma) \\ &= H F^t \mathbf{x}'' + \sum_{\sigma=0}^{t-1} H F^{t-1-\sigma} G \mathbf{u}(\sigma) = \mathbf{y}''(t) \quad t = 0, 1, \dots, k-1. \end{aligned} \quad (7.20)$$

$\mathbf{x}'$ and $\mathbf{x}''$ are indistinguishable (notation: $\mathbf{x}' \sim \mathbf{x}''$) if (7.20) holds true for every $k \geq 0$.

It is easy to verify that $\mathbf{x}' \underset{[0, k-1]}{\sim} \mathbf{x}''$ if and only if $\mathbf{x}'$ and $\mathbf{x}''$ induce the same unforced output evolution at times $0, 1, \dots, k-1$, and hence if and only if $\mathbf{x}' - \mathbf{x}''$ is in $X_{[0, k-1]}^{\text{no}}$.

$\mathbf{x}'$ and $\mathbf{x}''$ are indistinguishable when they induce the same unforced output evolutions in $[0, +\infty)$ and this is the case if and only if $\mathbf{x}' - \mathbf{x}''$ belongs to the non-observable space X^{no} .

¹The indistinguishability notion here considered is occasionally referred to in the literature as “indistinguishability in the future” and opposed to the “indistinguishability in the past”, that we will investigate later.

Indistinguishability in $[0, k-1]$ and indistinguishability are equivalence relations in X (i.e., they enjoy the reflexivity, symmetry and transitivity properties). In both cases, the state-space X is partitioned into equivalence classes endowed with a very simple structure: as far as indistinguishability in $[0, k-1]$ is concerned, the class including vector $\mathbf{x}$ consists of all the vectors taking the following form

$$\mathbf{x} + \mathbf{v}, \quad \mathbf{v} \in X_{[0, k-1]}^{\text{no}},$$

and is denoted by $\mathbf{x} + X_{[0, k-1]}^{\text{no}}$; for indistinguishability, on the other hand, the class including vector $\mathbf{x}$ consists of all the vectors

$$\mathbf{x} + \mathbf{v}, \quad \mathbf{v} \in X^{\text{no}},$$

and is denoted by $\mathbf{x} + X^{\text{no}}$.

If the system is observable (observable in $[0, k-1]$), X^{no} ($X_{[0, k-1]}^{\text{no}}$) coincides with $\{\mathbf{0}\}$ and each indistinguishability class (indistinguishability class in $[0, k-1]$) includes exactly one state. Two different initial states then induce different unforced output evolutions (unforced output evolutions that differ in the interval $[0, k-1]$), so that - as previously remarked - the initial state is uniquely determined by the input and output data.

If the system is not observable (not observable in $[0, k-1]$), each equivalence class includes infinite vectors. From the input and output data (up to time $k-1$) one can identify the equivalence class of the initial state, but not the specific initial state within the class. In other words, one can identify only the initial state up to a vector in X^{no} (in $X_{[0, k-1]}^{\text{no}}$): no information is available that allows to single out a specific vector in the class.

Remark 3 If $X_{[0, k-1]}^{\text{no}}$ has dimension greater than or equal to 1 and the unforced output is available (either by direct measurement or as a result of the knowledge of the input and output data), equation (7.9) can be solved, but the solution is not unique.

The general solution, obtained as the sum of a particular solution $\bar{\mathbf{x}}(0)$ of (7.9) and an arbitrary vector in $\ker \mathcal{O}_{[0, k-1]} = X_{[0, k-1]}^{\text{no}}$, is the indistinguishability class in $[0, k-1]$ $\bar{\mathbf{x}}(0) + X_{[0, k-1]}^{\text{no}}$ and it represents the set of all initial states that are compatible with the unforced output evolution.

Example 7.1.2 By referring to the system of Example 7.1.1, assume $\lambda_1 = \lambda_2 = \lambda$ and $h_1 = h_2 = 1$. The non-observable subspace (i.e., the non-observable subspace in $[0, 1]$) is

$$X^{\text{no}} = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 = -x_2 \right\}$$

and the equivalence classes are the lines parallel to the line $x_1 = -x_2$.

The states $\mathbf{x}'$ and $\mathbf{x}''$ are indistinguishable, while the states $\mathbf{x}'$ and $\mathbf{x}'''$ can be distinguished.

Starting from the unforced output evolution, one can deduce which line of $\mathbb{R}^2$ the initial state belongs to, but one cannot uniquely determine the specific position on the line. For instance, if the output is

$$y_\ell(0) = 2, \quad y_\ell(1) = 2\lambda,$$

the initial state is any solution of the equation

$$\begin{bmatrix} 2 \\ 2\lambda \end{bmatrix} = \begin{bmatrix} H \\ HF \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \lambda & \lambda \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}.$$

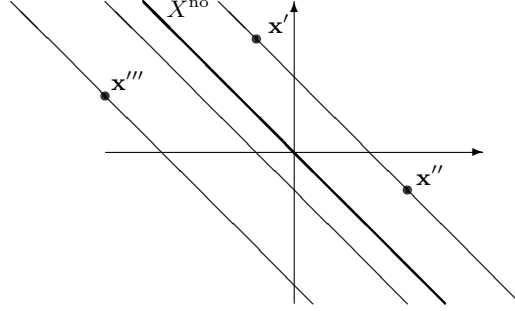


Figure 7.1.2

A specific solution is $\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ while the general solution is expressed as

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} + X_2^{\text{no}} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} \alpha \\ -\alpha \end{bmatrix}, \alpha \in \mathbb{R} \right\}.$$

7.2 Reconstructibility of linear discrete-time systems

Suppose now to know the input samples $\mathbf{u}(0), \mathbf{u}(1), \dots, \mathbf{u}(k-2)$ and the output samples $\mathbf{y}(0), \mathbf{y}(1), \dots, \mathbf{y}(k-1)$ and to be interested in determining the system state at the final time $k-1$.

If the system is observable in $[0, k-1]$, the problem admits a unique solution, since the inputs and the outputs in the interval $[0, k-1]$ uniquely determine $\mathbf{x}(0)$ and hence, by making use of

$$\mathbf{x}(k-1) = F^{k-1}\mathbf{x}(0) + \sum_{\sigma=0}^{k-2} F^{k-2-\sigma} G \mathbf{u}(\sigma), \quad (7.21)$$

one can determine the state at $k-1$.

If the system is not observable in $[0, k-1]$, the input and output sequences do not uniquely determine the initial state, but only the indistinguishability class $\mathbf{x}(0) + X_{[0, k-1]}^{\text{no}}$ to which it belongs. In other words, one can only claim that if $\mathbf{x}(0)$ is an initial state that is compatible with the data, then the states of the class $\mathbf{x}(0) + X_{[0, k-1]}^{\text{no}}$ are all the states (and the only ones) that are compatible with the data. Based on (7.21), the admissible states at time $k-1$ are the states that can be obtained starting at $t=0$ from any element of the class $\mathbf{x}(0) + X_{[0, k-1]}^{\text{no}}$ and applying the assigned input sequence. They generate the class

$$F^{k-1}\mathbf{x}(0) + F^{k-1}X_k^{\text{no}} + \sum_{\sigma=0}^{k-2} F^{k-2-\sigma} G \mathbf{u}(\sigma). \quad (7.22)$$

The class (7.22) includes a single element, and hence the state at time $k-1$ is uniquely determined, if and only if the *non-reconstructible subspace* in $[0, k-1]$ $F^{k-1}X_k^{\text{no}}$ only includes the zero vector, namely if

$$\ker F^{k-1} \supseteq X_k^{\text{no}}. \quad (7.23)$$

Condition (7.23) represents a necessary and sufficient condition to be able to uniquely determine the final state at $t = k-1$ from the input and output data on the interval

$[0, k-1]$. When so, the system is said to be *reconstructable in $[0, k-1]$* . It is immediate to verify that:

1. if a discrete-time system is observable in $[0, k-1]$, it is reconstructable in $[0, k-1]$;
2. reconstructability in $[0, k-1]$ does not ensure observability in $[0, k-1]$;
3. if a system is reconstructable in $[0, k-1]$, it is also reconstructable in $[0, k]$.

- EXERCISE 7.2.1 Provide an example of a discrete-time system that is reconstructable in $[0, 1]$, but not observable in $[0, k-1]$, for any k .
- EXERCISE 7.2.2 [NON-RECONSTRUCTABLE SUBSPACE] Prove that the *non-reconstructable subspace in $[0, k-1]$* $X_{[0, k-1]}^{nr} := F^{k-1} X_{[0, k-1]}^{no}$, $k \geq 1$,
 - i) Includes all the vectors (and those only) that may be the system state at time $t = k-1$ when the unforced output evolution is zero in $[0, k-1]$;
 - ii) $X_{[0, k-1]}^{nr} \supseteq X_{[0, k]}^{nr}$;
 - iii) If $X_{[0, k-1]}^{nr} = X_{[0, k]}^{nr}$ then $X_{[0, k]}^{nr} = X_{[0, k+1]}^{nr}$.

‡ Hint for iii): Saying that $\mathbf{x} \in X_{[0, k]}^{nr}$ amounts to assuming that $\mathbf{x}$ takes the form $\mathbf{x} = F^k \mathbf{z}$, for some $\mathbf{z} \in X_{[0, k]}^{no}$. Since $\tilde{\mathbf{x}} := F^{k-1} \mathbf{z}$ belongs to $X_{[0, k-1]}^{nr}$, the hypothesis in iii) implies $\tilde{\mathbf{x}} \in X_{[0, k]}^{nr}$. Then there exists $\tilde{\mathbf{z}} \in X_{[0, k]}^{no}$ such that $\tilde{\mathbf{x}} = F^k \tilde{\mathbf{z}}$. From $F\tilde{\mathbf{x}} = F^k \mathbf{z} = \mathbf{x}$ it follows $\mathbf{x} = F^{k+1} \tilde{\mathbf{z}}$ and from $HF^{k+1} \tilde{\mathbf{z}} = HF^k \mathbf{z} = 0$ it follows that $\tilde{\mathbf{z}} \in X_{[0, k+1]}^{no}$. Consequently, $\mathbf{x} \in X_{[0, k+1]}^{nr}$.

System $\Sigma = (F, G, H)$ is said to be *reconstructable* if it is reconstructable in $[0, k-1]$ for some $k \in \mathbb{N}$, namely if from the knowledge of the input and output data on a sufficiently large time interval, it is possible to uniquely determine the final state.

Proposition 7.2.1 Let $\Sigma = (F, G, H)$ be a discrete-time state-space model of dimension n . The following facts are equivalent:

i) Σ is reconstructable;

ii)

$$\ker F^n \supseteq \ker \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix} = X^{no}; \quad (7.24)$$

iii) Σ is reconstructable in $[0, n]$.

PROOF i) $\Rightarrow$ ii) We preliminarily observe (see Proposition A.2.3) that for every positive integer k one has

$$\ker F^n \supseteq \ker F^k. \quad (7.25)$$

Keeping into account (7.6) and (7.25), if the system is reconstructable in $[0, k-1]$ for some k , one has $\ker F^{k-1} \supseteq X_{[0, k-1]}^{no}$ and hence

$$\ker F^n \supseteq \ker F^{k-1} \supseteq X_{[0, k-1]}^{no} \supseteq X_{[0, n-1]}^{no} = X^{no} \quad (7.26)$$

which proves (7.24).

ii) $\Rightarrow$ iii) From (7.24) it follows $\ker F^n \supseteq X_{[0, n-1]}^{no} = X_{[0, n]}^{no}$, which is a necessary and sufficient condition for the reconstructability in $[0, n]$.

iii) $\Rightarrow$ i) It is obvious from the definition of reconstructable system. ■

- **EXERCISE 7.2.3** Consider a system $\Sigma = (F, G, H)$ of dimension n , and assume that F is a nilpotent Jordan block with nilpotency index equal to n and $H = 0$. The system is reconstructable in $n + 1$ steps, but not in n .

Example 7.2.1 Consider the following model of the dynamics of an age-structured population, with three age classes, the last of them not being reproductive,

$$\begin{aligned}x_1(t+1) &= \alpha_1 x_1(t) + \alpha_2 x_2(t), \\x_2(t+1) &= \beta_1 x_1(t), \\x_3(t+1) &= \beta_2 x_2(t),\end{aligned}$$

where α_i , $i = 1, 2$, are the fertility rates and β_i , $i = 1, 2$, the survival rates. Suppose to measure as system output the number of newborn individuals in every time interval: $y(t) = x_1(t)$. Then, the system matrices (assuming lack of migration phenomena, and hence unforced evolution) are

$$F = \begin{bmatrix} \alpha_1 & \alpha_2 & 0 \\ \beta_1 & 0 & 0 \\ 0 & \beta_2 & 0 \end{bmatrix}, \quad H = [1 \quad 0 \quad 0].$$

Let us consider the following problems:

P.1 Determine the initial population distribution $\mathbf{x}(0)$ by making use of the data about the newborn individuals in the time intervals from $t = 0$ onward (i.e., the output data $y(0), y(1), \dots$);

P.2 Determine the final population distribution $\mathbf{x}(0)$ by making use of the data about the newborn individuals in the time intervals up to $t = 0$ (i.e., the output data $y(0), y(-1), \dots$).

P.1 is an observability problem, while P.2 a reconstructability one. As far as observability is concerned, one has

$$\mathcal{O} = \begin{bmatrix} 1 & 0 & 0 \\ \alpha_1 & \alpha_2 & 0 \\ \alpha_1^2 + \alpha_2\beta_1 & \alpha_1\alpha_2 & 0 \end{bmatrix}. \quad (7.27)$$

If we suppose that α_2 is positive, namely that the second age class is reproductive, (8.9) has rank 2 and the system is not observable, with non-observable subspace of dimension 1. Since not all initial population distributions $\mathbf{x}(0)$, can be distinguished from the zero population, problem P.1 cannot be solved. More specifically, all the populations distributions whose unique nonzero class is the one that is no longer reproductive, namely all the population vectors of the following type

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix}, \quad x_3 \in \mathbb{R}_+,$$

are indistinguishable from the zero population.

As far as reconstructability is concerned, one has

$$\ker F^3 \supseteq \ker \mathcal{O}.$$

Therefore the system is reconstructable and P.2 is solvable.

One can directly verify that P.1 cannot be solved, by explicitly writing the system of equations in the unknown variables $x_i(0)$, $i = 1, 2, 3$. From

$$\begin{aligned}y(0) &= H\mathbf{x}(0) = x_1(0) \\y(1) &= HF\mathbf{x}(0) = \alpha_1 x_1(0) + \alpha_2 x_2(0) \\y(2) &= HF^2\mathbf{x}(0) = (\alpha_1^2 + \alpha_2\beta_1)x_1(0) + \alpha_1\alpha_2 x_2(0)\end{aligned}$$

one verifies that $x_1(0)$ and $x_2(0)$ can be uniquely determined, while $x_3(0)$ does not appear in the equations and hence can take an arbitrary value. $x_3(0)$ cannot be determined from the data about newborn individuals from $t = 0$ onward, because it represents the number of individuals belonging to the no longer reproductive class, that has no influence on the future output, which is just the

number of individuals in the first class.

As far as problem P.2 is concerned, the available data are $y(0), y(-1), y(-2), \dots$. From equation

$$\mathbf{x}(0) = F\mathbf{x}(-1) = \begin{bmatrix} \alpha_1 & \alpha_2 & 0 \\ \beta_1 & 0 & 0 \\ 0 & \beta_2 & 0 \end{bmatrix} \begin{bmatrix} x_1(-1) \\ x_2(-1) \\ x_3(-1) \end{bmatrix} = \begin{bmatrix} \alpha_1 x_1(-1) + \alpha_2 x_2(-1) \\ \beta_1 x_1(-1) \\ \beta_2 x_2(-1) \end{bmatrix},$$

keeping in mind that $y(t) = x_1(t)$, one has

$$\begin{aligned} y(0) = x_1(0) &= \alpha_1 x_1(-1) + \alpha_2 x_2(-1) = \alpha_1 y(-1) + \alpha_2 x_2(-1), \\ x_2(0) &= \beta_1 x_1(-1) = \beta_1 y(-1), \\ x_3(0) &= \beta_2 x_2(-1). \end{aligned}$$

Upon deducing $x_2(-1)$ from the first equation and by replacing it in the third one, one finally obtains

$$\begin{aligned} x_1(0) &= y(0) \\ x_2(0) &= \beta_1 y(-1) \\ x_3(0) &= \frac{\beta_2}{\alpha_2} y(0) - \frac{\alpha_1 \beta_2}{\alpha_2} y(-1), \end{aligned}$$

which shows that the final state is reconstructable (in two steps).

7.3 Observability and reconstructability of linear continuous-time systems

Let us consider, now, some methods to study observability and reconstructability of a continuous-time linear system, of dimension n ,

$$\begin{aligned} \dot{\mathbf{x}}(t) &= F\mathbf{x}(t) + G\mathbf{u}(t), \\ \mathbf{y}(t) &= H\mathbf{x}(t). \end{aligned} \tag{7.28}$$

A state $\mathbf{x}$ is not observable in the interval $[0, t]$, with $t > 0$, if, for every input function $\mathbf{u}(\cdot)$, the output $\mathbf{y}(\cdot)$ corresponding to $\mathbf{u}(\cdot)$ and to the initial state $\mathbf{x}$ coincides in $[0, t]$ with the output corresponding to $\mathbf{u}(\cdot)$ and the initial state $\mathbf{0}$. Keeping into account that the forced output evolutions are the same in both cases, the state $\mathbf{x}$ is not observable in $[0, t]$ if and only if

$$He^{F\sigma}\mathbf{x} = 0, \quad 0 \leq \sigma \leq t. \tag{7.29}$$

If we denote by ω_t the linear map

$$\omega_t : X \rightarrow C[0, t]^p : \mathbf{x} \mapsto He^{F\sigma}\mathbf{x}, \quad 0 \leq \sigma \leq t, \tag{7.30}$$

that associates with the initial state $\mathbf{x}$ the unforced output function² that it induces in the interval $[0, t]$, it is clear that $\mathbf{x}$ is not observable in $[0, t]$ if and only if $\mathbf{x} \in \ker \omega_t$. The set of non-observable states in $[0, t]$ is therefore a subspace

$$X_{[0, t]}^{\text{no}} := \ker \omega_t \tag{7.31}$$

of the state space X , that we will refer to as *non-observable subspace* in $[0, t]$.

² $C[0, t]^p$ denotes the space of all functions that are continuous in $[0, t]$ and take values in $\mathbb{R}^p$.

The following proposition allows to express in a very simple way the non-observable subspace $X_{[0,t]}^{\text{no}}$ of a continuous-time system as a matrix kernel. Since this latter does not depend on t , it follows that $X_{[0,t]}^{\text{no}}$ is independent of the length t of the interval where input and output evolutions are known, as far as $t > 0$.

Proposition 7.3.1 *The non-observable subspace $X_{[0,t]}^{\text{no}}$ of system (7.28) coincides, for every $t > 0$, with the kernel of the observability matrix*

$$\mathcal{O} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix}. \quad (7.32)$$

PROOF $\mathbf{x}$ is non-observable in $[0, t]$ if and only if $\mathbf{y}_\ell(\sigma) = He^{F\sigma}\mathbf{x} = 0$, $\forall \sigma \in [0, t]$, namely if and only if

$$\sum_{k=0}^{\infty} \frac{HF^k \mathbf{x}}{k!} \sigma^k = 0, \quad \forall \sigma \in [0, t]. \quad (7.33)$$

By applying the identity principle of the power series and Cayley-Hamilton theorem, condition (7.33) is equivalent to $HF^k \mathbf{x} = 0$, $k = 0, 1, 2, \dots, n-1$, and hence to

$$\mathbf{x} \in \ker \mathcal{O}. \quad (7.34)$$

■

As the subspace $X_{[0,t]}^{\text{no}} = \ker \mathcal{O}$ does not depend on t , it will be simply denoted by X^{no} and referred to as *non-observable subspace*, without additional specifications.

If $X^{\text{no}} = \{\mathbf{0}\}$, namely if $\mathcal{O}$ is of full rank n , we will say that system (7.28) is *observable*. In this case the system unforced output evolution can be identically zero in a right neighbourhood of $t = 0$ only if the initial state is the zero state.

- EXERCISE 7.3.1 Suppose that system (7.28) is observable and it evolves, without the action of forcing inputs, starting from the state $\mathbf{x}(0) = \bar{\mathbf{x}}$.
 - i) If the output is zero in some interval $[\tau_1, \tau_2]$, with $0 \leq \tau_1 < \tau_2$, then $\bar{\mathbf{x}} = \mathbf{0}$.
 - ii) If for every $j = 1, 2, \dots, p$ the j -th component of the output, $y_j(\cdot)$, becomes zero in $[\tau_{1j}, \tau_{2j}]$, with $0 \leq \tau_{1j} < \tau_{2j}$, then $\bar{\mathbf{x}} = \mathbf{0}$.

‡ Hint for ii). Denote by $\mathcal{O}^{(j)}$ the observability matrix of the pair $(F, \mathbf{e}_j^T H)$, for $j = 1, 2, \dots, p$. By following the procedure of Proposition 7.3.1 verify that $\mathcal{O}^{(j)} \bar{\mathbf{x}} = \mathbf{0}$, $\forall j$. Therefore $\mathcal{O} \bar{\mathbf{x}} = \mathbf{0}$.
- EXERCISE 7.3.2 The non-observable subspace, both for discrete-time and for continuous-time systems, is F -invariant and can be characterised as the largest F -invariant subspace included in $\ker H$.

‡ Hint. As far as F -invariance is concerned, verify that $\mathbf{x} \in \ker \mathcal{O} \Rightarrow F\mathbf{x} \in \ker \mathcal{O}$; the inclusion $\ker \mathcal{O} \subseteq \ker H$ is obvious. Every F -invariant subspace $V \subseteq \ker H$ is also included in $\ker(HF^k)$, $\forall k > 0$.

Example 7.3.1 In the electrical network of Figure 7.3.1 assume the applied voltage as input $u(\cdot)$ and the current absorbed by the circuit as output $y(\cdot)$.

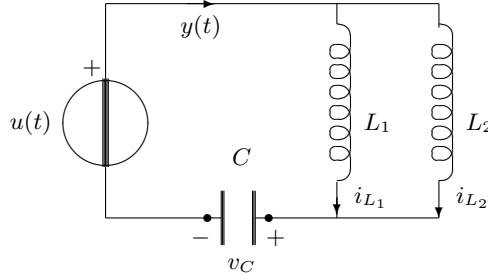


Figure 7.3.1

the inductors i_{L_1} and i_{L_2} and the voltage drop along the capacitor v_C , respectively, the state equations are

$$\begin{aligned}\dot{x}_1(t) &= \frac{1}{L_1}(u(t) - x_3(t)) \\ \dot{x}_2(t) &= \frac{1}{L_2}(u(t) - x_3(t)) \\ \dot{x}_3(t) &= \frac{1}{C}(x_1(t) + x_2(t)) \\ y(t) &= x_1(t) + x_2(t)\end{aligned}$$

and they identify the matrices $F, \mathbf{g}, H$ given by

$$F = \begin{bmatrix} 0 & 0 & -\frac{1}{L_1} \\ 0 & 0 & -\frac{1}{L_2} \\ \frac{1}{C} & \frac{1}{C} & 0 \end{bmatrix}, \quad \mathbf{g} = \begin{bmatrix} \frac{1}{L_1} \\ \frac{1}{L_2} \\ 0 \end{bmatrix}, \quad H = [1 \quad 1 \quad 0].$$

The observability matrix

$$\mathcal{O} = \begin{bmatrix} H \\ HF \\ HF^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & -(\frac{1}{L_1} + \frac{1}{L_2}) \\ -\frac{1}{C}(\frac{1}{L_1} + \frac{1}{L_2}) & -\frac{1}{C}(\frac{1}{L_1} + \frac{1}{L_2}) & 0 \end{bmatrix}$$

has rank 2. Therefore, starting from the recorded data about u and y it is not possible to determine the triple of initial values of x_1, x_2, x_3 .

The non-observable subspace is given by

$$X^{\text{no}} = \ker \mathcal{O} = \{\mathbf{x} : x_1 + x_2 = 0, x_3 = 0\} = \left\{ \begin{bmatrix} \alpha \\ -\alpha \\ 0 \end{bmatrix}, \alpha \in \mathbb{R} \right\},$$

and two initial states $\mathbf{x}'$ and $\mathbf{x}''$ are indistinguishable if $\mathbf{x}' - \mathbf{x}'' \in \ker \mathcal{O}$ namely if

$$\mathbf{x}' - \mathbf{x}'' = \begin{bmatrix} \alpha \\ -\alpha \\ 0 \end{bmatrix}$$

for some real α .

The obtained result admits a very simple physical interpretation: the possible presence of an initial current circulating in the loop consisting of L_1 and L_2 does not influence the circuit behaviour as it is seen from the external terminals.

By resorting to the adjoint operator of ω_t

$$\omega_t^* : C[0, t]^p \rightarrow X : \mathbf{y}(\cdot) \mapsto \int_0^t e^{F^T \sigma} H^T \mathbf{y}(\sigma) d\sigma, \quad (7.35)$$

we can obtain an additional characterisation of the non-observable subspace $X_{[0, t]}^{\text{no}}$. Indeed, from the properties of the adjoint operator

Suppose that one can record, in a certain time interval, both the applied voltage and the absorbed current. We want to understand whether this information allows to determine the initial values of the currents in the inductors and of the voltage drops along the capacitors.

This is clearly an observability problem. To answer this question, one can write the state equations of the electrical network and evaluate the observability matrix.

If we denote by x_1, x_2 and x_3 the currents in

$$\ker \omega_t = \ker(\omega_t^* \omega_t) = \ker \int_0^t e^{F^T \sigma} H^T H e^{F \sigma} d\sigma, \quad (7.36)$$

it follows that $X_{[0,t]}^{\text{no}}$ is the kernel of the $n \times n$ semipositive definite matrix

$$\mathcal{V}_t := \int_0^t e^{F^T \sigma} H^T H e^{F \sigma} d\sigma, \quad (7.37)$$

known as *observability gramian corresponding to the interval* $[0, t]$. In particular, it turns out that $X_{[0,t]}^{\text{no}} = \{\mathbf{0}\}$, i.e., the system is observable, if and only if the gramian has rank n . If the unforced output evolution $\mathbf{y}_\ell(\cdot)$ in the interval $[0, t]$ is available, from

$$\mathbf{y}_\ell(\cdot) = \omega_t \mathbf{x}(0) \quad (7.38)$$

it follows that

$$\omega_t^* \mathbf{y}_\ell(\cdot) = \omega_t^* \omega_t \mathbf{x}(0) = \mathcal{V}_t \mathbf{x}(0) \quad (7.39)$$

and hence, if the system is observable,

$$\mathbf{x}(0) = \mathcal{V}_t^{-1} \omega_t^* \mathbf{y}_\ell(\cdot) = \left[\int_0^t e^{F^T \sigma} H^T H e^{F \sigma} d\sigma \right]^{-1} \int_0^t e^{F^T \sigma} H^T \mathbf{y}_\ell(\sigma) d\sigma. \quad (7.40)$$

Remark 1 If system (7.28) is not observable and $\mathbf{y}_\ell(\cdot)$ is one of its unforced output evolutions in the time interval $[0, t]$, equation (7.38) in the unknown variable $\mathbf{x}(0)$ has a solution (since $\mathbf{y}_\ell(\cdot) \in \text{Im } \omega_t$), but the solution is not unique (because $\dim \ker \omega_t > 0$). Instead of (7.38) we can consider (7.39), i.e.,

$$\omega_t^* \mathbf{y}_\ell(\cdot) = \int_0^t e^{F^T \sigma} H^T \mathbf{y}_\ell(\sigma) d\sigma = \mathcal{V}_t \mathbf{x}(0).$$

It has the structure of a system of n linear equations in n unknown variables, and it admits the same solutions as (7.38). Indeed, every solution of (7.38) is trivially a solution of (7.39). On the other hand, if $\mathbf{x}(0)$ is a solution of (7.39), it satisfies $\mathbf{0} = \mathcal{V}_t \mathbf{x}(0) - \omega_t^* \mathbf{y}_\ell(\cdot) = \omega_t^* (\omega_t \mathbf{x}(0) - \mathbf{y}_\ell(\cdot))$, and hence $(\omega_t \mathbf{x}(0) - \mathbf{y}_\ell(\cdot)) \in \ker \omega_t^* \cap \text{Im } \omega_t = (\text{Im } \omega_t)^\perp \cap \text{Im } \omega_t = \{\mathbf{0}\}$. Therefore $\mathbf{x}(0)$ is a solution of (7.38).

Remark 2 If the system is observable and the measurable output $\tilde{\mathbf{y}}_\ell(\cdot)$ is affected by noise, by proceeding as in the discrete-time case we can show that the initial state given by (7.40) is the one that generates the unforced output evolution that is the closest in norm to $\tilde{\mathbf{y}}_\ell(\cdot)$. In the non-observable case, this same statement can be made for each of the states $\mathbf{x}(0)$ that solve (7.39). The details are left as an exercise.

- EXERCISE 7.3.3 For the continuous-time system $\Sigma = (F, -, H)$, any pair of the following statements implies the third one: (i) Σ is observable; (ii) the Lyapunov equation $F^T X + X F = -H^T H$ admits a positive definite solution; (iii) $\text{Re } \Lambda(F) < 0$, namely each eigenvalue of F has negative real part.
- EXERCISE 7.3.4 If the continuous-time system $\Sigma = (F, -, H)$ is observable and internally stable, the energy of the unforced output evolution on the infinite horizon $[0, +\infty)$ is finite for every initial state $\mathbf{x}(0)$ and is expressed by the quadratic form $\|\mathbf{y}_\ell\|_2^2 = \int_{t=0}^\infty \mathbf{y}_\ell^T(t) \mathbf{y}_\ell(t) dt = \mathbf{x}(0)^T \int_{t=0}^\infty e^{F^T t} H^T H e^{F t} dt \mathbf{x}(0)$.

The matrix $\mathcal{V}_\infty := \int_{t=0}^\infty e^{F^T t} H^T H e^{F t} dt$ is the only positive definite solution of the Lyapunov equation $F^T X + X F = -H^T H$.

By a reasoning analogous to the one adopted for discrete-time systems, one can introduce also for a continuous-time system the *indistinguishability relation between two states $\mathbf{x}'$ and $\mathbf{x}''$* . It reduces to the identity, on $[0, +\infty)$ or on every non-degenerate interval $[0, t]$, of the unforced output evolutions induced by the two states, and it amounts to condition $\mathbf{x}' - \mathbf{x}'' \in X^{\text{no}}$.

From the input/output data it is possible to identify the equivalence class $\mathbf{x}(0) + X^{\text{no}}$ to which the system state at $t = 0$ belongs, but not the state itself, except for the case when the system is observable. At the generic time instant $t > 0$ all the states belonging to the set

$$e^{Ft}\mathbf{x}(0) + e^{Ft}X^{\text{no}} + \int_0^t e^{F(t-\sigma)}G\mathbf{u}(\sigma)d\sigma \quad (7.41)$$

are therefore compatible with the input/output data. So, the system state at time t is uniquely determined if and only if $e^{Ft}X^{\text{no}} = \{\mathbf{0}\}$. By the invertibility of the exponential matrix this happens if and only if $X^{\text{no}} = \{\mathbf{0}\}$ and the conclusion is that a continuous-time system is reconstructable (on any non-degenerate interval $[0, t]$) if and only if the system is observable.

7.4 Duality

The study of reachability and controllability, as well as that of observability and reconstructability, has been carried out along lines that exhibit some form of similarity, both in the definitions and in the corresponding characterisations. This correspondence can be formally justified by resorting to the notion of dual system.

Given a continuous-time or discrete-time system $\Sigma = (F, G, H)$, system $\Sigma_d := (F^T, H^T, G^T)$ is called the *dual system* of Σ . The number of inputs and outputs of Σ_d is equal to the number of outputs and inputs of Σ , respectively.

It is immediate to see that, if $\mathcal{R}_d$ and $\mathcal{O}_d$ denote the reachability and observability matrices of Σ_d , respectively, they are related to the matrices $\mathcal{R}$ and $\mathcal{O}$ of Σ by the relationships

$$\mathcal{R}_d = [H^T \quad F^T H^T \quad \dots \quad (F^T)^{n-1} H^T] = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{n-1} \end{bmatrix}^T = \mathcal{O}^T, \quad (7.42)$$

and, analogously,

$$\mathcal{O}_d = \mathcal{R}^T. \quad (7.43)$$

For discrete-time systems, Σ is controllable if and only if condition

$$\text{Im}F^n \subseteq \text{Im}\mathcal{R}$$

holds, which is equivalent to $(\text{Im}F^n)^\perp \supseteq (\text{Im}\mathcal{R})^\perp$, and hence to

$$\ker(F^n)^T = \ker(F^T)^n \supseteq \ker\mathcal{R}^T = \ker\mathcal{O}_d$$

which is the reconstructability condition for Σ_d . Analogously, one can see that Σ is reconstructable if and only if Σ_d is controllable.

Finally, note that the dual of system Σ_d is system Σ .

The relationships expressed by the previous formulas are summarised in the following table:

Σ		Σ_d
reachable	$\Leftrightarrow$	observable
observable	$\Leftrightarrow$	reachable
controllable	$\Leftrightarrow$	reconstructable
reconstructable	$\Leftrightarrow$	controllable

The notion of dual system is extremely useful from a practical viewpoint, because it allows to obtain criteria and canonical forms for observability from criteria and canonical forms derived for reachability.

1. [ALGEBRAIC EQUIVALENCE] Given two algebraically equivalent systems

$$\Sigma = (F, G, H) \quad , \quad \bar{\Sigma} = (P^{-1}FP, P^{-1}G, HP),$$

the corresponding duals are given by

$$\Sigma_d = (F^T, H^T, G^T) \quad , \quad \bar{\Sigma}_d = (P^T F^T (P^T)^{-1}, P^T H^T, G^T (P^T)^{-1})$$

and they are in turn algebraically equivalent:

$\Sigma = (F, G, H) \quad \xrightarrow{P} \quad \bar{\Sigma} = (P^{-1}FP, P^{-1}G, HP)$
$\updownarrow \text{ dual.}$
$\Sigma_d = (F^T, H^T, G^T) \quad \xrightarrow{(P^T)^{-1}} \quad \bar{\Sigma}_d = (P^T F^T (P^T)^{-1}, P^T H^T, G^T (P^T)^{-1})$

If the change of basis matrix from Σ to $\bar{\Sigma}$ is P , the one from Σ_d to $\bar{\Sigma}_d$ is $(P^T)^{-1}$.

2. [OBSERVABILITY STANDARD FORM] Let $\Sigma = (F, G, H)$ be a non-observable system. Then $\Sigma_d = (F^T, H^T, G^T)$ is non-reachable and there exists a change of basis matrix Q that reduces it to reachability standard form:

$$\bar{\Sigma}_d = (Q^{-1}F^TQ, Q^{-1}H^T, G^TQ) = \left(\begin{bmatrix} \bar{F}_{11}^T & \bar{F}_{21}^T \\ 0 & \bar{F}_{22}^T \end{bmatrix}, \begin{bmatrix} \bar{H}_1^T \\ 0 \end{bmatrix}, [\bar{G}_1^T \quad \bar{G}_2^T] \right), \quad (7.44)$$

where $(\bar{F}_{11}^T, \bar{H}_1^T)$ is a reachable pair. In the dual system $\bar{\Sigma}$ of $\bar{\Sigma}_d$, described by

$$\bar{\Sigma} = (\bar{F}, \bar{G}, \bar{H}) = \left(\begin{bmatrix} \bar{F}_{11} & 0 \\ \bar{F}_{21} & \bar{F}_{22} \end{bmatrix}, \begin{bmatrix} \bar{G}_1 \\ \bar{G}_2 \end{bmatrix}, [\bar{H}_1 \quad 0] \right), \quad (7.45)$$

the pair $(\bar{F}_{11}, \bar{H}_1)$ is observable, because it is obtained by duality from a reachable pair. $\bar{\Sigma}$ represents the *observability standard form* of system Σ , to which Σ is algebraically equivalent, with change of basis matrix $P = (Q^T)^{-1}$. The situation is described in the following diagram:

$$\begin{array}{ccc}
\Sigma = (F, G, H) & \xrightarrow{P=(Q^T)^{-1}} & \bar{\Sigma} = \left(\begin{bmatrix} \bar{F}_{11} & 0 \\ \bar{F}_{21} & \bar{F}_{22} \end{bmatrix}, \begin{bmatrix} \bar{G}_1 \\ \bar{G}_2 \end{bmatrix}, [\bar{H}_1 \quad 0] \right) \\
\updownarrow \text{dual.} & & \updownarrow \text{dual.} \\
\Sigma_d = (F^T, H^T, G^T) & \xrightarrow{Q} & \bar{\Sigma}_d = \left(\begin{bmatrix} \bar{F}_{11}^T & \bar{F}_{21}^T \\ 0 & \bar{F}_{22}^T \end{bmatrix}, \begin{bmatrix} \bar{H}_1^T \\ 0 \end{bmatrix}, [\bar{G}_1^T \quad \bar{G}_2^T] \right)
\end{array}$$

Matrix P , that realises the change of basis that leads to the observability standard form for the non-observable system, is here obtained from the matrix Q that is responsible, for the dual system, of the transition to the reachability standard form. Its inverse P^{-1} can be obtained directly from the observability matrix of Σ , without involving the dual system: indeed, the first columns of Q represent a maximal set of linearly independent columns of $\mathcal{R}_d$, and therefore the first rows of $Q^T = P^{-1}$ are a maximal set of linearly independent rows of $\mathcal{R}_d^T = \mathcal{O}$.

3. [OBSERVABLE AND NON-OBSERVABLE SUB-SYSTEMS] If we denote by $\bar{\mathbf{x}} = P^{-1}\mathbf{x}$ the n -tuple that represents the state vector of system (7.45) and we partition it conformably to the partition of the matrices $(\bar{F}, \bar{G}, \bar{H})$, we obtain the equations

$$\begin{aligned}
\bar{\mathbf{x}}_1(t+1) &= \bar{F}_{11}\bar{\mathbf{x}}_1(t) + \bar{G}_1\mathbf{u}(t) \\
\bar{\mathbf{x}}_2(t+1) &= \bar{F}_{21}\bar{\mathbf{x}}_1(t) + \bar{F}_{22}\bar{\mathbf{x}}_2(t) + \bar{G}_2\mathbf{u}(t) \\
\mathbf{y}(t) &= \bar{H}_1\bar{\mathbf{x}}_1(t)
\end{aligned}$$

corresponding to the block-scheme of Figure 7.4.1.

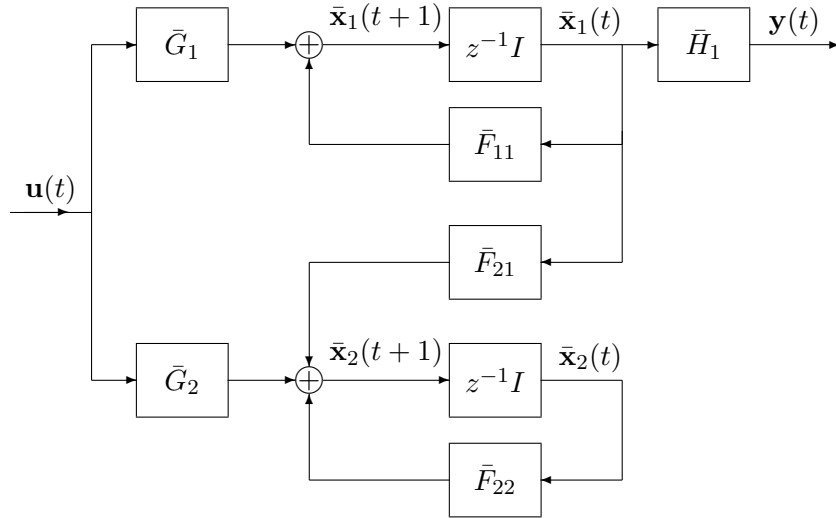


Figure 7.4.1

Subsystem $(\bar{F}_{11}, \bar{G}_1, \bar{H}_1)$ is called *observable subsystem* of system (7.45). As the pair

$(\bar{F}_{11}, \bar{H}_1)$ is observable, the state of this subsystem can be determined on the basis of the input and output data.

The non-observable subspace X^{no} of system (7.45) consists of all (and only those) vectors whose block $\bar{\mathbf{x}}_1$ is zero:

$$X^{\text{no}} = \left\{ \begin{bmatrix} \bar{\mathbf{x}}_1 \\ \bar{\mathbf{x}}_2 \end{bmatrix} : \bar{\mathbf{x}}_1 = \mathbf{0} \right\}.$$

Indeed, they are the only state vectors determining an unforced output evolution for system (7.45) which is identically zero. Subsystem $(\bar{F}_{22}, \bar{G}_2, 0)$ is called the *non-observable subsystem*.

4. [TRANSFER MATRIX] It is immediate to verify that the system transfer matrix (7.45), identical to the one of the algebraically equivalent system $\Sigma = (F, G, H)$, coincides in turn with the transfer matrix of the observable subsystem:

$$W(z) = H(zI - F)^{-1}G = \bar{H}_1(zI - \bar{F}_{11})^{-1}\bar{G}_1.$$

This result is also obvious from Figure 7.4.1: for every input $\mathbf{u}(\cdot)$, the corresponding output is determined only by the observable subsystem and, when its initial conditions are zero, simply by the transfer matrix of such a subsystem.

5. [PBH OBSERVABILITY TEST] System $\Sigma = (F, G, H)$ is observable if and only if the polynomial matrix

$$\begin{bmatrix} F - zI_n \\ H \end{bmatrix} \quad (7.46)$$

is of full rank for every $z \in \mathbb{C}$, or, equivalently, for every z belonging to the spectrum of F . The proof is straightforward and simply requires to apply the PBH reachability test to the dual system. So, it is left as an exercise.

6. [OBSERVABLE CANONICAL FORM FOR SINGLE-OUTPUT SYSTEMS] Every single-output observable system $\Sigma = (F, G, H)$ is algebraically equivalent to a system of the following form

$$\begin{aligned} F_o = Q^{-1}FQ &= \begin{bmatrix} 0 & 0 & & -\alpha_0 \\ 1 & 0 & & -\alpha_1 \\ & 1 & \ddots & -\alpha_2 \\ & & \ddots & \vdots \\ & & & 1 & -\alpha_{n-1} \end{bmatrix}, & G_o = Q^{-1}G, \\ H_o = HQ &= [0 \quad 0 \quad \cdots \quad 0 \quad 1], \end{aligned} \quad (7.47)$$

where $\alpha_0 + \alpha_1 z + \cdots + \alpha_{n-1} z^{n-1} + z^n$ is the characteristic polynomial of F . To verify (7.47) it is sufficient to reduce the dual system Σ_d to the controllable canonical form. The matrix Q is given by $(P^{-1})^T$, where P is the change of basis matrix that reduces Σ_d to controllable canonical form. Matrix Q can be obtained directly from the observability matrices $\mathcal{O}$ of Σ and $\mathcal{O}_o$ of (7.47): indeed, $\mathcal{O}_o = \mathcal{O}Q$, from which

$$Q = \mathcal{O}^{-1}\mathcal{O}_o. \quad (7.48)$$

7. [MULTIVARIABLE OBSERVABLE CANONICAL FORM] By a procedure similar to the one of the previous point, one can obtain the *multivariable observable canonical form* for

an observable system Σ with several outputs. It is sufficient to reduce the corresponding (reachable) dual system Σ_d to multivariable controllable canonical form $\Sigma_d^{(\text{can})}$ and construct the dual of $\Sigma_d^{(\text{can})}$. The structure of the multivariable observable canonical form depends on the *observability Kronecker indices*, that can be defined in terms of the observable pair (F, H) in the same way as the control invariants introduced in the previous chapter.

- EXERCISE 7.4.1 Given a system $\Sigma = (F, G, H)$ and its dual $\Sigma_d = (F^T, H^T, G^T)$, the transfer matrices $W(z)$ and $W_d(z)$ satisfy $W_d(z) = W^T(z)$. In particular, if two SISO systems are one the dual of the other, they share the same transfer function.
- EXERCISE 7.4.2 [LYAPUNOV EQUATION AND OBSERVABILITY] Keeping in mind Lemma 5.5.1, solve Exercise 7.1.4 by resorting to duality: for the discrete-time system $\Sigma = (F, -, H)$, any pair of the following three statements imply the third one: (i) Σ is observable; (ii) the Lyapunov equation $F^T X F - X = -H^T H$ admits a positive definite solution; (iii) $|\Lambda(F)| < 1$.

7.5 Canonical decomposition

Consider system $\Sigma = (F, G, H)$ and let X^R and X^{no} denote its reachable and non-observable subspaces, respectively. As they are both F -invariant, their intersection $X_2 := X^R \cap X^{\text{no}}$ is, in turn, F -invariant.

Let $\mathcal{B}_2$ be a basis for X_2 , and let $\mathcal{B}_1$ and $\mathcal{B}_4$ be two sets of linearly independent vectors such that $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for X^R , and $\mathcal{B}_2 \cup \mathcal{B}_4$ a basis for X^{no} . Finally, let $\mathcal{B}_3$ be a set of linearly independent vectors such that by juxtaposing the vectors of $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{B}_3$ and $\mathcal{B}_4$ we obtain a basis for X . With respect to that basis, matrices F, G and H take the form

$$\begin{aligned} F &= \begin{bmatrix} F_{11} & 0 & F_{13} & 0 \\ F_{21} & F_{22} & F_{23} & F_{24} \\ 0 & 0 & F_{33} & 0 \\ 0 & 0 & F_{43} & F_{44} \end{bmatrix}, \quad G = \begin{bmatrix} G_1 \\ G_2 \\ 0 \\ 0 \end{bmatrix}, \\ H &= [H_1 \quad 0 \quad H_3 \quad 0]. \end{aligned} \quad (7.49)$$

To obtain (11.81) it is sufficient to remember that $X^R = \text{span}\{\mathcal{B}_1, \mathcal{B}_2\}$ and $X^{\text{no}} = \text{span}\{\mathcal{B}_2, \mathcal{B}_4\}$ are F -invariant, that $\text{Im } G \subseteq X^R$ and that $\ker H \supseteq X^{\text{no}}$.

From the structure of (11.81) one can determine the following subsystems:

$$\Sigma^{(\text{RO})} = (F_{11}, G_1, H_1) \quad \text{reachable and observable subsystem;}$$

$$\Sigma^{(\text{R})} = \left(\begin{bmatrix} F_{11} & 0 \\ F_{21} & F_{22} \end{bmatrix}, \begin{bmatrix} G_1 \\ G_2 \end{bmatrix}, [H_1 \quad 0] \right) \quad \text{reachable subsystem;}$$

$$\Sigma^{(\text{O})} = \left(\begin{bmatrix} F_{11} & F_{13} \\ 0 & F_{33} \end{bmatrix}, \begin{bmatrix} G_1 \\ 0 \end{bmatrix}, [H_1 \quad H_3] \right) \quad \text{observable subsystem;}$$

$$\Sigma^{(\text{NR,NO})} = (F_{44}, 0, 0) \quad \text{non-reachable and non-observable subsystem.}$$

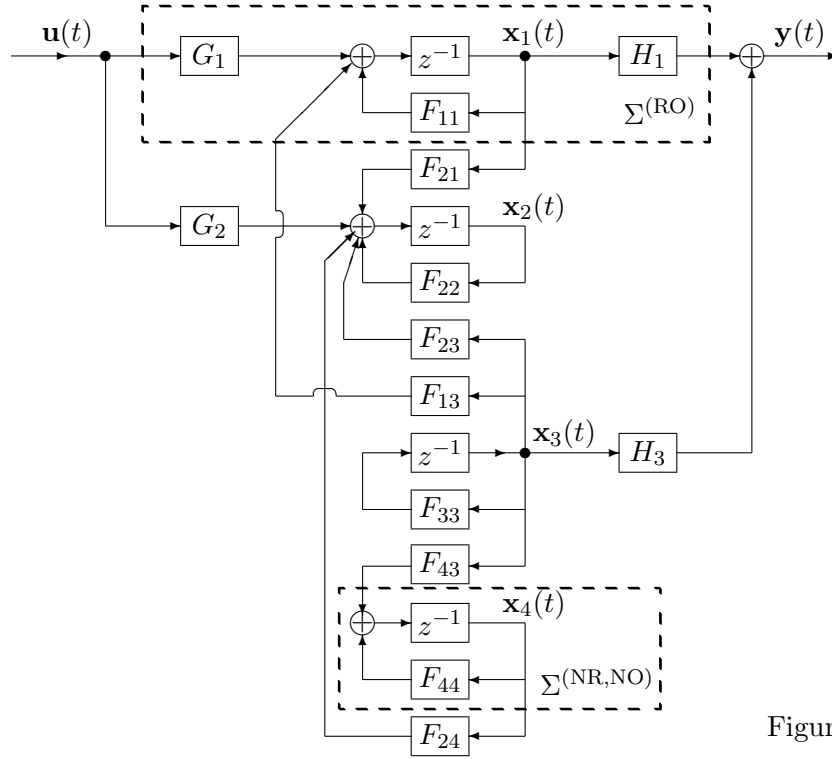


Figure 7.5.1

Obviously, the transfer matrix only depends on the reachable and observable subsystem $\Sigma^{(RO)}$, and hence:

$$W(z) = H(zI - F)^{-1}G = H_1(zI - F_{11})^{-1}G_1.$$

By referring to the basis we have adopted

$$\underbrace{\mathcal{B}_1 \cup \mathcal{B}_2}_{\text{basis for reachable subsp.}} \cup \overbrace{\mathcal{B}_4}^{\text{basis for non-obs. subsp.}} \cup \mathcal{B}_3$$

- the vectors of subspace $\langle \mathcal{B}_2 \rangle$ are those (and those only) that are simultaneously reachable and not observable, namely the states of X that can be reached from the zero state and that, if assumed as initial states, generate identically zero unforced output evolutions;

- the vectors of subspace $\langle \mathcal{B}_4 \rangle$ are those (and those only) that are simultaneously not reachable and not observable, namely the states of X that cannot be reached from the zero state and that, if assumed as initial states, generate identically zero unforced output evolutions;

- the vectors of subspace $\langle \mathcal{B}_3 \rangle$ are those (and those only) that are simultaneously not reachable and observable, namely the states of X that cannot be reached from the zero state but that, if assumed as initial states, generate non-identically zero unforced output evolutions.

Remark The fact that $\Sigma^{(R)}$ is reachable does not imply that the subsystem $(F_{22}, G_2, 0)$ is completely reachable. In other words, as an isolated subsystem it represents the “reachable and non-observable subsystem”: starting from the zero state, in the overall system all the states of the subspace generated by $\mathcal{B}_2$ are reachable, but this in general happens thanks also to the connection with subsystem $\Sigma^{(RO)}$ via block F_{21} .

Analogously, the observability of $\Sigma^{(O)}$ does not imply that the isolated subsystem $(F_{33}, 0, H_3)$ is observable and not reachable. From the unforced output evolution data, it is possible to determine the components with respect to $\mathcal{B}_3$ of the system initial state, but in general this happens thanks to the connection realised through F_{13} .

For instance, in the (sub)system

$$\Sigma^{(O)} = \left(\begin{bmatrix} F_{11} & F_{13} \\ 0 & F_{33} \end{bmatrix}, \begin{bmatrix} G_1 \\ 0 \end{bmatrix}, [H_1 \quad H_3] \right) = \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, [1 \quad 0] \right)$$

the pair (F_{33}, H_3) is non-observable, even if system $\Sigma^{(O)}$ is observable. If we consider the unforced evolution, the output of block H_3 is identically zero independently of the initial state $\mathbf{x}_3(0)$, but the connection F_{13} allows nonetheless to determine, through the output of block H_1 , all the entries of the initial state of $\Sigma^{(O)}$, and in particular $\mathbf{x}_3(0)$.

7.6 Asymptotic state-observers

Several control techniques, including eigenvalue allocation, are based on the assumption that all state variables are available for measurement at every time instant. Oftentimes, this is either impossible or not convenient: only part of the state variables can be measured at every time instant, and hence only a partial measure of the internal dynamics of the system is available. In these situations, two possible strategies can be chosen: one can either resort to complex control strategies that are based on the dynamic elaboration of the measured state variables, or one tries to obtain from the measured (= output) variables the information about the remaining state variables, and then use this information for control.

This latter method is based on the introduction of a dynamical system, known as state-observer, that provides an estimate of the state vector based on the available data. In this case, the feedback control strategies may later be implemented by making use of the state estimate, rather than the real state vector.

7.6.1 Full-order observers

The simplest state-observer is the so-called *open-loop observer*, which is obtained by creating a copy of the system $\Sigma = (F, G, -)$. By referring to a discrete-time system

$$\begin{aligned} \mathbf{x}(t+1) &= F\mathbf{x}(t) + G\mathbf{u}(t), \\ \mathbf{y}(t) &= H\mathbf{x}(t), \end{aligned}$$

(the procedure for a continuous-time system is analogous), the open-loop observer equation is

$$\hat{\mathbf{x}}(t+1) = F\hat{\mathbf{x}}(t) + G\mathbf{u}(t). \quad (7.50)$$

Upon defining the estimation error at time t as

$$\mathbf{e}(t) = \hat{\mathbf{x}}(t) - \mathbf{x}(t), \quad (7.51)$$

one gets

$$\mathbf{e}(t+1) = \hat{\mathbf{x}}(t+1) - \mathbf{x}(t+1) = F(\hat{\mathbf{x}}(t) - \mathbf{x}(t)) = F\mathbf{e}(t).$$

Starting from an estimate $\hat{\mathbf{x}}(0)$ of the initial state $\mathbf{x}(0)$, associated to which we have the initial estimation error $\mathbf{e}(0) = \hat{\mathbf{x}}(0) - \mathbf{x}(0)$, the error dynamics $\mathbf{e}(t) = F^t\mathbf{e}(0)$ depends on the eigenvalues of F .

So, for instance, if system Σ is unstable, the estimation error, instead of converging to zero as t grows, can diverge. On the other hand, even when Σ is asymptotically stable, it may be impossible to guarantee the error convergence to zero with the desired rate, unless the system stability margin is sufficiently high.

Open-loop observers do not take advantage of one of the available data: the output evolution $\mathbf{y}(\cdot)$ of Σ . On the contrary, this is used in the observer scheme appearing in Figure 7.6.2, to produce a feedback control law that is based on the comparison between the output $H\mathbf{x}(t)$ of system Σ and the quantity $H\hat{\mathbf{x}}(t)$ (a sort of estimated output). This kind of observer is called *closed-loop observer* and since its dimension is the same as the dimension of the original system Σ , it is also called *full-order closed-loop observer*.

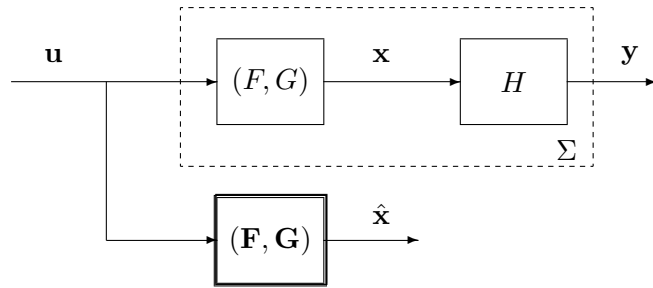


Figure 7.6.1

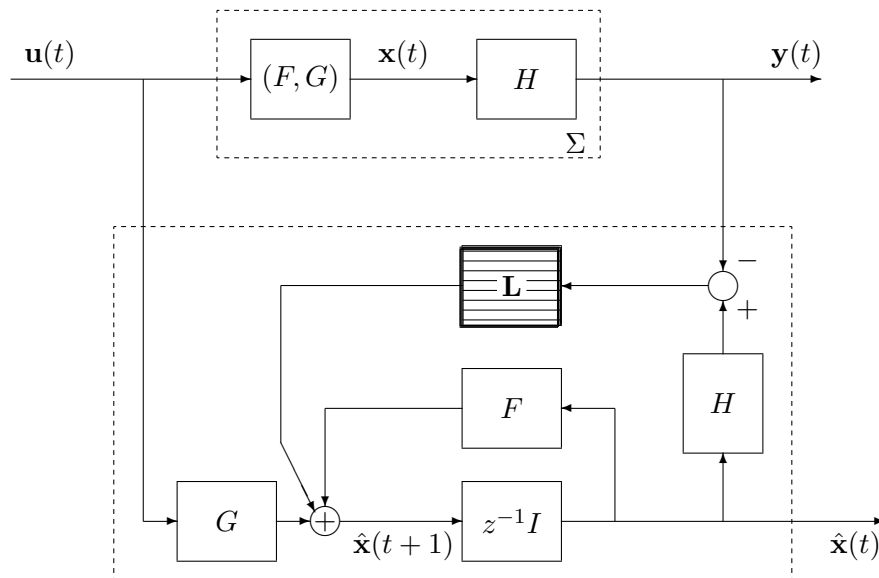


Figure 7.6.2

The block with the horizontal lines is a static element described by an $n \times p$ matrix L , that we can arbitrarily choose in order to influence the estimation error dynamics. The full-order closed-loop observer equations are:

$$\hat{\mathbf{x}}(t+1) = F\hat{\mathbf{x}}(t) + L(H\hat{\mathbf{x}}(t) - \mathbf{y}(t)) + G\mathbf{u}(t),$$

namely

$$\hat{\mathbf{x}}(t+1) = (F + LH)\hat{\mathbf{x}}(t) - L\mathbf{y}(t) + G\mathbf{u}(t). \quad (7.52)$$

The estimation error now satisfies the equation

$$\begin{aligned} \mathbf{e}(t+1) &= \hat{\mathbf{x}}(t+1) - \mathbf{x}(t+1) \\ &= (F + LH)\hat{\mathbf{x}}(t) - LH\mathbf{x}(t) + G\mathbf{u}(t) - F\mathbf{x}(t) - G\mathbf{u}(t) \\ &= (F + LH)\mathbf{e}(t). \end{aligned} \quad (7.53)$$

In this case, the estimation error dynamics is determined by the eigenvalues of matrix $F + LH$, and we can influence their location in the complex plane, by suitably choosing L .

How much freedom we have in allocating the eigenvalues of $F + LH$ by acting on L depends on the structural properties of the pair (F, H) . More specifically, being

$$\det(zI - (F + LH)) = \det(zI - (F + LH))^T = \det(zI - (F^T + H^T L^T)), \quad (7.54)$$

we see that as L varies within the set of all $n \times p$ real matrices, the characteristic polynomial of $F + LH$ can arbitrarily vary within the set of all monic polynomials of degree n if and only if the pair (F^T, H^T) is reachable and hence if and only if the pair (F, H) is observable. Once we have arbitrarily chosen the eigenvalues of $F + LH$, under the assumption that (F, H) is observable, equation (7.54) suggests a procedure to determine L . Indeed, consider the reachable pair (F^T, H^T) and, by proceeding according to the techniques illustrated in the previous chapters, one can determine a state-feedback control matrix K in order to allocate the eigenvalues of $F^T + H^T K$. Then, we will just need to assume $L = K^T$.

If (F, H) is not observable, the error dynamics matrix $F + LH$ has some fixed eigenvalues, that cannot be influenced by the choice of L . As one can easily verify by reducing the system to standard observability form, those eigenvalues (and elementary modes) are those of the non-observable subsystem. Consequently, a (*full-order closed-loop*) *asymptotic state-observer* exists if and only if the *non-observable subsystem is asymptotically stable*. A system whose non-observable subsystem is asymptotically stable is called *detectable*.

Example 7.6.1 Consider the continuous-time system $\Sigma = (F, \mathbf{g}, H)$, described by the equations

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 1 \\ -1 & 0 & -1 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) = F\mathbf{x}(t) + \mathbf{g}u(t) \\ y(t) &= [1 \quad 0 \quad 0] \mathbf{x}(t) = H\mathbf{x}(t). \end{aligned}$$

We want to design an asymptotic state-observer having all the eigenvalues equal to -3 .

- By following, step by step, what we previously said, we know that if Σ is observable, then its dual $\Sigma_d = (F^T, H^T, \mathbf{g}^T)$ is reachable and there exists a matrix $K = [k_0 \quad k_1 \quad k_2]$ such that

$$\Delta_{F^T+H^TK}(s) = (s+3)^3.$$

To determine K , one can adopt the procedure described in Chapter 6: the controllability canonical form of Σ_d is given by the pair

$$(F^T)_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}, \quad (H^T)_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad (7.55)$$

and, upon denoting by $\mathcal{R}_d$ and by $\mathcal{R}_{dc}$ the reachability matrices of (F^T, H^T) and $((F^T)_c, (H^T)_c)$, respectively, the change of basis matrix to obtain the controllable canonical form is

$$T = \mathcal{R}_d \mathcal{R}_{dc}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

The matrix K_c such that

$$\det(sI - (F^T)_c - (H^T)_c K_c) = (s+3)^3 = s^3 + 9s^2 + 27s + 27 \quad (7.56)$$

can be immediately derived from (9.36) and (7.56):

$$K_c = [-26 \quad -25 \quad -6]$$

and from it we derive

$$K = K_c T^{-1} = [-6 \quad -7 \quad -7].$$

The observer matrix is then

$$L = \begin{bmatrix} \ell_0 \\ \ell_1 \\ \ell_2 \end{bmatrix} = K^T = \begin{bmatrix} -6 \\ -7 \\ -7 \end{bmatrix}. \quad (7.57)$$

- Alternatively, one can directly proceed by imposing that the characteristic polynomial

$$\begin{aligned} \det(sI_3 - F - LH) &= \det \begin{bmatrix} s - \ell_0 & -1 & 0 \\ -\ell_1 & s + 2 & -1 \\ 1 - \ell_2 & 0 & s + 1 \end{bmatrix} \\ &= s^3 + (3 - \ell_0)s^2 + (2 - 3\ell_0 - \ell_1)s + (1 - 2\ell_0 - \ell_1 - \ell_2) \end{aligned}$$

coincides with $(s+3)^2 = s^3 + 9s^2 + 27s + 27$. This leads to the solution of a system of linear equations in the three unknown variables ℓ_0, ℓ_1, ℓ_2 , and (7.57) is immediately obtained.

When dealing with discrete-time systems, we have seen that (F, H) is reconstructable if and only if (F^T, H^T) is controllable. In this case it is possible to allocate all the eigenvalues of $F + LH$ in the origin, by obtaining a so-called *dead-beat observer*, whose estimation error goes to zero in a finite number of steps.

- EXERCISE 7.6.1 Prove that for a discrete-time system $\Sigma = (F, G, H)$ the following properties are equivalent:
 - Σ is reconstructable;
 - the non-observable subsystem of Σ has all zero eigenvalues;
 - Σ admits a dead-beat observer;
 - matrix $\begin{bmatrix} F - zI \\ H \end{bmatrix}$ has full rank for every nonzero $z \in \mathbb{C}$.

7.6.2 Reduced-order observers

Full-order observers, that we have discussed up to now, have a certain redundancy. Indeed, they produce an estimate of all n state variables, by neglecting the fact that output variables are related to state variables via a static relationship, depending on the entries of matrix H . The idea at the basis of reduced-order observers consists in taking advantage of the relationship between output and state variables, in order to extract from the output in a “static way” as much information about the state as possible, thus leaving to the observer, which is a dynamic systems, the task of providing an estimate of the remaining state variables.

1. The static part of the procedure is immediate and it relies on a suitable change of basis in the state-space X .

Suppose that the p rows of H , namely the p outputs of the system, are linearly independent. This amounts to assuming

$$\text{rank} H = p.$$

If H is not of full row rank, one can refer to the matrix H' , of full row rank, that is obtained by selecting in H a maximal number of linearly independent rows. In this way, the output components that are neglected, being linearly dependent on the others, do not provide additional information about the state.

Let us construct an $n \times n$ change of basis matrix T by imposing that the rows of H are the last p rows of T^{-1} , namely

$$T^{-1} = \left[\begin{array}{c} V \\ H \end{array} \right] \left\{ \begin{array}{l} n-p \\ p \end{array} \right\} \quad (7.58)$$

and choosing the $n-p$ rows of submatrix V so that T^{-1} is non-singular. With respect to the new basis, the state is represented by the vector

$$\bar{\mathbf{x}} = T^{-1}\mathbf{x} = \left[\begin{array}{c} V\mathbf{x} \\ H\mathbf{x} \end{array} \right] = \left[\begin{array}{c} \mathbf{w} \\ \mathbf{y} \end{array} \right] \left\{ \begin{array}{l} n-p \\ p \end{array} \right\} \quad (7.59)$$

The change of basis introduced in X allows to identify in the last p state variables the system outputs, and hence the only state variables that need to be estimated are the $n-p$ components of the vector $\mathbf{w}$.

With respect to the new basis the system matrices are

$$A := T^{-1}FT = \left[\begin{array}{cc} A_{11} & A_{12} \\ A_{21} & A_{22} \end{array} \right], \quad B := T^{-1}G = \left[\begin{array}{c} B_1 \\ B_2 \end{array} \right], \quad [0 \quad I_p] = HT, \quad (7.60)$$

and the state-update equations can be expressed in the form

$$\begin{aligned} \mathbf{w}(t+1) &= A_{11}\mathbf{w}(t) + A_{12}\mathbf{y}(t) + B_1\mathbf{u}(t), \\ \mathbf{y}(t+1) &= A_{21}\mathbf{w}(t) + A_{22}\mathbf{y}(t) + B_2\mathbf{u}(t). \end{aligned} \quad (7.61)$$

2. Let us denote by L an $(n-p) \times p$ matrix, whose choice will be discussed later, and let us assume

$$\mathbf{v}(t) := \mathbf{w}(t) + L\mathbf{y}(t). \quad (7.62)$$

By premultiplying the second equation in (7.61) by L and summing it up to the first equation, we obtain

$$\mathbf{v}(t+1) = (A_{11} + LA_{21})\mathbf{w}(t) + (A_{12} + LA_{22})\mathbf{y}(t) + (B_1 + LB_2)\mathbf{u}(t); \quad (7.63)$$

from which, by adding and subtracting to the right hand side of (7.63) the term $(A_{11} + LA_{21})L\mathbf{y}$, one deduces the update equation of $\mathbf{v}(t)$

$$\mathbf{v}(t+1) = (A_{11} + LA_{21})\mathbf{v}(t) + (A_{12} + LA_{22} - A_{11}L - LA_{21}L)\mathbf{y}(t) + (B_1 + LB_2)\mathbf{u}(t), \quad (7.64)$$

which is the state-update equation of a system with state vector $\mathbf{v}$, dimension equal to $n - p$, and inputs $\mathbf{u}$ and $\mathbf{y}$.

3. Consider an open-loop observer, of order $n - p$, for the state of system (7.64):

$$\hat{\mathbf{v}}(t+1) = (A_{11} + LA_{21})\hat{\mathbf{v}}(t) + (A_{12} + LA_{22} - A_{11}L - LA_{21}L)\mathbf{y}(t) + (B_1 + LB_2)\mathbf{u}(t). \quad (7.65)$$

The estimation error $\mathbf{e}_v(t) = \hat{\mathbf{v}}(t) - \mathbf{v}(t)$ satisfies equation

$$\mathbf{e}_v(t+1) = \hat{\mathbf{v}}(t+1) - \mathbf{v}(t+1) = (A_{11} + LA_{21})\mathbf{e}_v(t),$$

and its dynamic is determined by the eigenvalues of $A_{11} + LA_{21}$. These eigenvalues can be arbitrarily assigned by resorting to a suitable choice of L if and only if the pair (A_{11}, A_{21}) is observable.

It turns out that the pair (A_{11}, A_{21}) is observable if and only if $(A, [0 \ I_p])$ is observable, and hence the (algebraically equivalent) pair (F, H) is observable. To verify this, it is convenient to remember that the observability of (F, H) is equivalent to the fact that the observability PBH matrix

$$\begin{bmatrix} A_{11} - zI_{n-p} & A_{12} \\ A_{21} & A_{22} - zI_p \\ 0 & I_p \end{bmatrix}, \quad (7.66)$$

is of full rank for every $z \in \mathbb{C}$. If this is the case, the first $n - p$ columns of (7.66) are linearly independent for every $z \in \mathbb{C}$, namely the matrix

$$\begin{bmatrix} A_{11} - zI_{n-p} \\ A_{21} \end{bmatrix}, \quad (7.67)$$

has full rank $(n - p)$ for every $z \in \mathbb{C}$, and this represents the observability condition for the pair (A_{11}, A_{21}) . Conversely, if matrix (7.67) has full rank for every $z \in \mathbb{C}$, it is easy to verify that the same can be said for (7.66).

4. To obtain an estimate $\hat{\mathbf{x}}(t)$ of the state $\mathbf{x}(t)$ of system (F, G, H) one can obtain the estimate $\hat{\mathbf{v}}(t)$ of $\mathbf{v}(t)$:

4.i) by assuming as estimate of $\mathbf{w}(t)$ the vector $\hat{\mathbf{w}}(t) = \hat{\mathbf{v}}(t) - L\mathbf{y}(t)$, the error in estimating $\mathbf{w}(t)$ is

$$\mathbf{e}_w(t) := \hat{\mathbf{w}}(t) - \mathbf{w}(t) = \mathbf{e}_v(t);$$

4.ii) the estimate $\hat{\bar{\mathbf{x}}}(t)$ of $\bar{\mathbf{x}}(t)$ is then

$$\begin{bmatrix} \hat{\mathbf{w}}(t) \\ \mathbf{y}(t) \end{bmatrix},$$

and the corresponding error is

$$\mathbf{e}_{\bar{\mathbf{x}}}(t) := \begin{bmatrix} \hat{\mathbf{w}}(t) \\ \mathbf{y}(t) \end{bmatrix} - \begin{bmatrix} \mathbf{w}(t) \\ \mathbf{y}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{e}_{\mathbf{v}}(t) \\ \mathbf{0} \end{bmatrix};$$

4.iii) by assuming as estimate of the state $\mathbf{x}(t) = T\bar{\mathbf{x}}(t)$ the vector $\hat{\mathbf{x}}(t) := T\hat{\bar{\mathbf{x}}}(t)$, the error is given by

$$\mathbf{e}_{\mathbf{x}}(t) := T(\hat{\bar{\mathbf{x}}}(t) - \bar{\mathbf{x}}(t)) = T\mathbf{e}_{\bar{\mathbf{x}}}(t) = T \begin{bmatrix} \mathbf{e}_{\mathbf{v}}(t) \\ \mathbf{0} \end{bmatrix} \quad (7.68)$$

and it converges to zero with the same rate of convergence as $\mathbf{e}_{\mathbf{v}}(t)$.

In Figure 7.6.3 the block scheme for the reduced-order observer we have just described is illustrated.

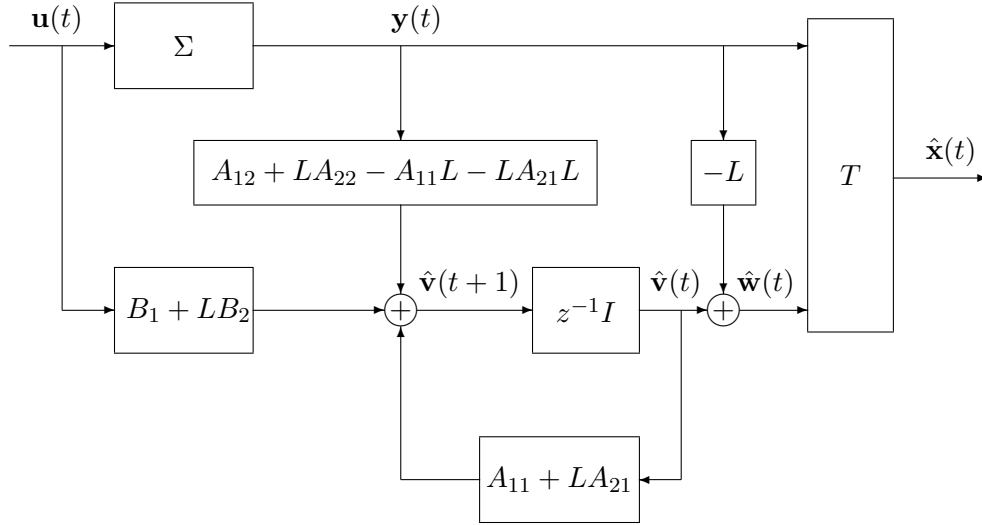


Figure 7.6.3

- EXERCISE 7.6.2 Prove that if (7.67) is of full rank for every $z \in \mathbb{C}$, then also (7.66) is of full rank for every $z \in \mathbb{C}$.
- EXERCISE 7.6.3 Given a system with m inputs $\Sigma = (F, G, H)$, assume that matrix G has rank m .
 - Determine a change of basis matrix T in the state-space so that

$$T^{-1}G = \begin{bmatrix} 0 \\ I_m \end{bmatrix}; \quad T^{-1}FT = A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}; \quad (7.69)$$

- show that the pair (F, G) is reachable if and only if the pair (A_{11}, A_{12}) is reachable;
- the eigenvalues of a square matrix A , block-partitioned as in (7.69), can be arbitrarily allocated via state-feedback by modifying the blocks A_{21} and A_{22} if and only if (A_{11}, A_{12}) is a reachable pair.

Example 7.6.2 Consider the system in Example 7.6.1. As $p = 1$, it is possible to construct a reduced-order observer, of dimension $n - p = 2$, with both eigenvalues equal to -3 . Assume

$$T^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

thus obtaining

$$\begin{aligned} A &= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \left[\begin{array}{cc|c} -1 & 0 & -1 \\ 1 & -2 & 0 \\ 0 & 1 & 0 \end{array} \right], \quad B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ - \end{bmatrix}, \\ C &= [0 \quad 0 \quad | \quad 1]. \end{aligned}$$

The observer design problem reduces to that of determining L so that

$$\det(sI - A_{11} - LA_{21}) = s^2 + 6s + 9,$$

namely

$$\det\left(\begin{bmatrix} s+1 & 0 \\ -1 & s+2 \end{bmatrix} - \begin{bmatrix} \ell_1 \\ \ell_2 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}\right) = s^2 + 6s + 9.$$

By solving the previous equation one obtains

$$\begin{bmatrix} \ell_1 \\ \ell_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$$

and the observer scheme is depicted in Figure 7.6.4.

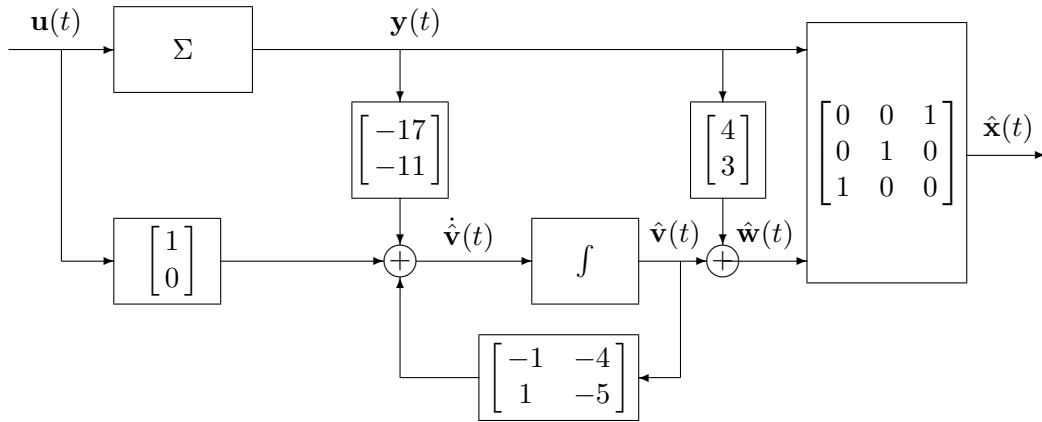


Figure 7.6.4

7.7 Design of an observer-based controller

Let us now investigate the dynamic properties of a system for which the feedback control law $\mathbf{u}(t) = K\mathbf{x}(t)$ is implemented by making use of the state estimate $\hat{\mathbf{x}}(t)$ provided by an observer, rather than the real state $\mathbf{x}(t)$.

In particular, we want to investigate the eigenvalue allocation problem for the overall system, and to determine its transfer matrix.

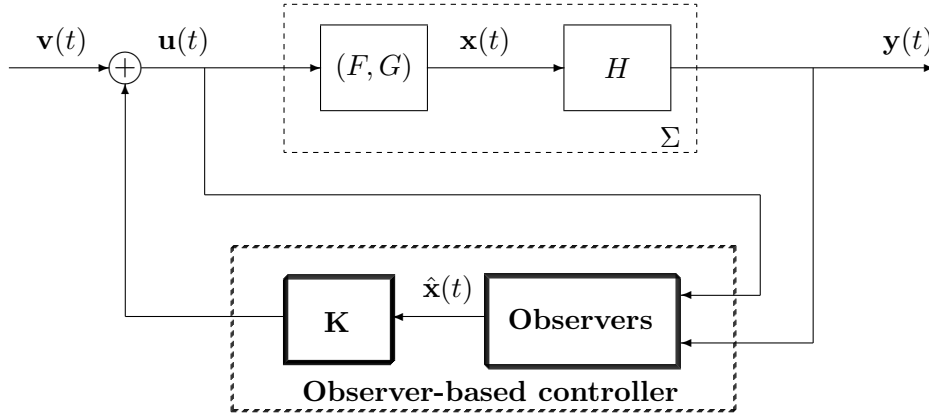


Figure 7.7.1

The scheme of Figure 7.7.1 represents the overall system obtained by connecting the original system Σ (*the plant*) with a full-order observer $\hat{\Sigma}$ and the static feedback K is applied to the state estimate.

The system obtained as a series of the observer and of the static element K is known as *observer-based controller*.

By referring to Figure 7.7.1, it is immediate to verify that the state space of the overall system is the direct sum of the state spaces of system Σ and of the observer $\hat{\Sigma}$, and to realise that its dynamics is described by the equations

$$\begin{aligned} \mathbf{x}(t+1) &= F\mathbf{x}(t) + G\mathbf{u}(t) \\ \mathbf{y}(t) &= H\mathbf{x}(t) \\ \mathbf{u}(t) &= \mathbf{v}(t) + K\hat{\mathbf{x}}(t) \\ \hat{\mathbf{x}}(t+1) &= F\hat{\mathbf{x}}(t) + G\mathbf{u}(t) + L(H\hat{\mathbf{x}}(t) - \mathbf{y}(t)) \end{aligned}$$

from which, after eliminating $\mathbf{u}(t)$, one obtains

$$\mathbf{x}(t+1) = F\mathbf{x}(t) + GK\hat{\mathbf{x}}(t) + G\mathbf{v}(t) \quad (7.70)$$

$$\hat{\mathbf{x}}(t+1) = -LH\mathbf{x}(t) + (F + LH + GK)\hat{\mathbf{x}}(t) + G\mathbf{v}(t) \quad (7.71)$$

$$\mathbf{y}(t) = H\mathbf{x}(t) \quad (7.72)$$

namely, in matrix form,

$$\begin{aligned} \begin{bmatrix} \mathbf{x}(t+1) \\ \hat{\mathbf{x}}(t+1) \end{bmatrix} &= \overbrace{\begin{bmatrix} F & | & GK \\ \hline -LH & | & F + LH + GK \end{bmatrix}}^A \begin{bmatrix} \mathbf{x}(t) \\ \hat{\mathbf{x}}(t) \end{bmatrix} + \overbrace{\begin{bmatrix} G \\ \hline G \end{bmatrix}}^B \mathbf{v}(t) \\ \mathbf{y}(t) &= \underbrace{\begin{bmatrix} H & | & 0 \end{bmatrix}}_C \begin{bmatrix} \mathbf{x}(t) \\ \hat{\mathbf{x}}(t) \end{bmatrix}. \end{aligned}$$

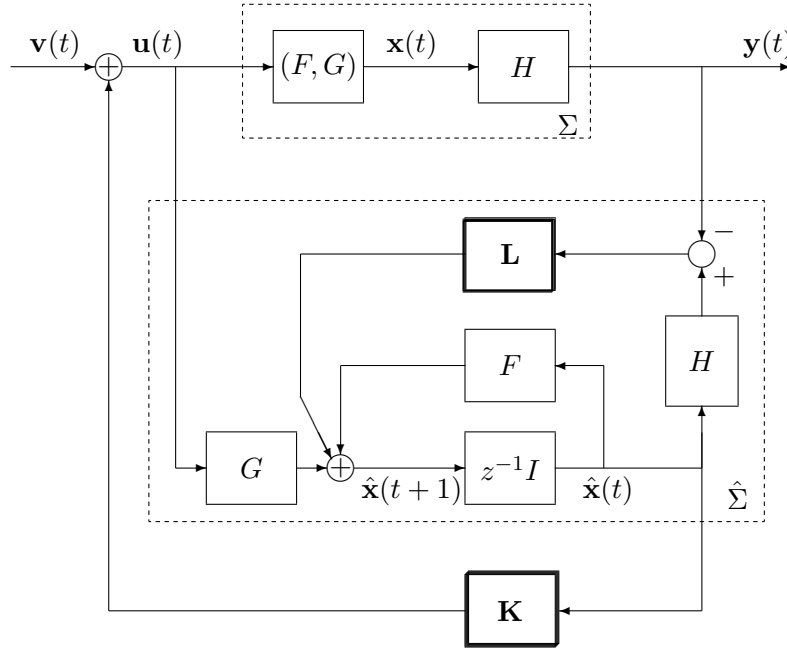


Figure 7.7.2

To answer the question we posed about the global system we have obtained, $\Sigma_{\text{global}} = (A, B, C)$ of dimension $2n$, it is necessary:

- (i) to evaluate the characteristic polynomial of matrix A ;
- (ii) to determine the input-output relationship of Σ_{global} .

For both (i) and (ii) it is convenient to introduce a change of basis in the state-space of the global system, associated with the matrix

$$T = \begin{bmatrix} I_n & 0 \\ I_n & I_n \end{bmatrix},$$

namely to assume as state of the global system the vector

$$T^{-1} \begin{bmatrix} \mathbf{x}(t) \\ \hat{\mathbf{x}}(t) \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ -I_n & I_n \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \hat{\mathbf{x}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{x}(t) \\ \hat{\mathbf{x}}(t) - \mathbf{x}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{e}(t) \end{bmatrix}. \quad (7.73)$$

The global system $\bar{\Sigma}_{\text{global}} = (\bar{A}, \bar{B}, \bar{C})$ one obtains in this way is algebraically equivalent to Σ_{global} and has matrices

$$\begin{aligned} \bar{A} &= T^{-1}AT = \begin{bmatrix} F + GK & GK \\ 0 & F + LH \end{bmatrix}, \\ \bar{B} &= T^{-1}B = \begin{bmatrix} G \\ 0 \end{bmatrix}, \\ \bar{C} &= CT = \begin{bmatrix} H & 0 \end{bmatrix}. \end{aligned} \quad (7.74)$$

It is then clear that the characteristic polynomial of $\bar{A}$, and hence of A , is given by

$$\begin{aligned}\det(zI_{2n} - \bar{A}) &= \det \begin{bmatrix} zI_n - F - GK & -GK \\ 0 & zI_n - F - LH \end{bmatrix} \\ &= \det(zI_n - F - GK) \det(zI_n - F - LH).\end{aligned}\quad (7.75)$$

This result, which is known in the literature as *separation property*, shows that when designing an observer-based controller the two basic steps, i.e.,

the synthesis of the state-feedback controller, namely the allocation of the eigenvalues of $F + GK$ by resorting to the state-feedback controller K ,

the observer design, namely the allocation of the eigenvalues of $F + LH$ by suitably choosing L ,

can be performed independently.

The spectrum of the global system, in which the feedback control acts on the state estimate, is simply the union of the spectra of $F + GK$ and $F + LH$.

It is then clear that the *stabilisation of the plant Σ by means of an observer-based controller is possible if and only if the non-reachable subsystem and the non-observable subsystem of Σ are both asymptotically stable, namely Σ is at the same time stabilisable and detectable.*

As far as the evaluation of the transfer matrix of Σ_{global} is concerned, by keeping into account the structures of the matrices in (10.135) we obtain

$$\begin{aligned}W_{\text{global}}(z) &= \bar{C}(zI_n - \bar{A})^{-1}\bar{B} \\ &= \begin{bmatrix} H & 0 \end{bmatrix} \begin{bmatrix} (zI_n - F - GK)^{-1} & * \\ 0 & (zI_n - F - LH)^{-1} \end{bmatrix} \begin{bmatrix} G \\ 0 \end{bmatrix} \\ &= H(zI_n - F - GK)^{-1}G.\end{aligned}$$

The transfer matrix we have obtained is the same one we would obtain by applying the state-feedback law directly to the real state variable $\mathbf{x}$, assuming it is accessible, without resorting to an observer. In other words, the observer does not affect the input/output relationship of the global system (assuming zero initial conditions).

The properties of the overall system that we have just described admit an immediate interpretation by referring to the scheme that describes the system with respect to the basis we have obtained after the change of basis associated with matrix T . Indeed, with respect to such a basis, the global state vector is given by (7.73), and the equations that describe the dynamics of $\mathbf{x}(t)$ and $\mathbf{e}(t)$ are

$$\begin{aligned}\mathbf{x}(t+1) &= (F + GK)\mathbf{x}(t) + GK\mathbf{e}(t) + G\mathbf{v}(t) \\ \mathbf{e}(t+1) &= (F + LH)\mathbf{e}(t) \\ \mathbf{y}(t) &= H\mathbf{x}(t),\end{aligned}$$

and correspond to the scheme in Figure 7.7.3.

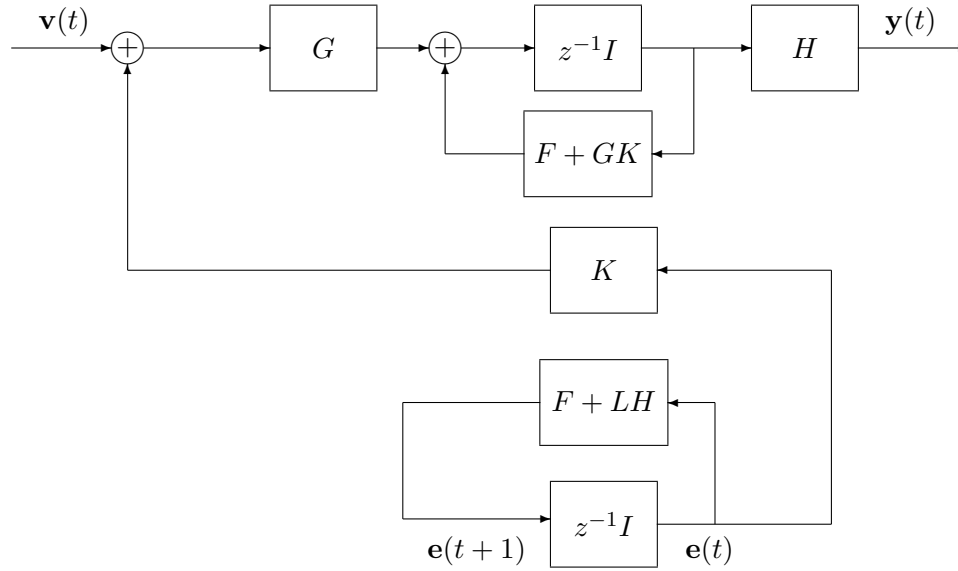


Figure 7.7.3

Clearly, the input-output relationship (when the initial state is zero) only depends on the subsystem $\Sigma_K = (F + GK, G, H)$. The estimation error $\mathbf{e}(t)$ is generated by a subsystem, evolving of pure unforced evolution, that produces an additional input $K\mathbf{e}(t)$ to system Σ_K , depending on the initial estimation error and asymptotically evolving to the zero state if the estimation error tends to zero.

Example 7.7.1 [CONTROL OF AN INVERTED PENDULUM]

A pole of mass m is hinged on a cart of mass $M \gg m$ whose movement is controlled by a force $u(\cdot)$.

The system equations are obtained by balancing the horizontal and the vertical components of the forces acting on the system and by balancing the moments of these forces. We neglect the effects of friction, and denote by $H(\cdot)$ and $V(\cdot)$ the horizontal and the vertical forces, respectively, exerted by the cart on the pole, by ℓ the distance of the pole center of mass from the point where the pole is hinged to the cart, and by J the moment of inertia of the pole with respect to his center of mass.

The motion of the center of mass of the pole is described by the equations

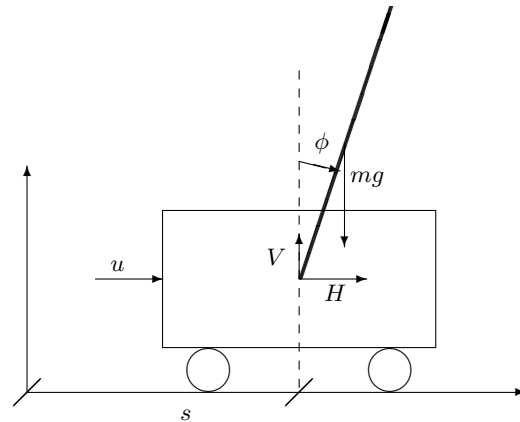


Figure 7.7.4

$$H(t) = m \frac{d^2}{dt^2} \left(s(t) + \ell \sin \phi(t) \right) \quad (7.76)$$

$$V(t) - mg = m \frac{d^2}{dt^2} \left(\ell \cos \phi(t) \right) \quad (7.77)$$

and the rotation movement with respect to the center of mass by

$$\ell \sin \phi(t) V(t) - \ell \cos \phi(t) H(t) = J \frac{d^2 \phi(t)}{dt^2}. \quad (7.78)$$

The cart motion obeys the equation

$$M \ddot{s}(t) = -H(t) + u(t)$$

that, by neglecting the reaction of the pole to the cart, becomes

$$M \ddot{s}(t) = u(t). \quad (7.79)$$

By developing the derivatives in (10.15) and (10.17) and by eliminating $V(t)$ and $H(t)$, one finally obtains the equation

$$(J + m\ell^2) \ddot{\phi} - mg\ell \sin \phi + m\ell \ddot{s} \cos \phi = 0. \quad (7.80)$$

Let us introduce the state variables

$$x_1 = \phi, \quad x_2 = \dot{\phi}$$

and let us use (7.79) to express $\ddot{s}$. We thus obtain the non-linear system

$$\begin{aligned} \dot{x}_1 &= x_2 & &:= f_1(x_1, x_2, u), \\ \dot{x}_2 &= \frac{gm\ell}{J + m\ell^2} \sin x_1 - \frac{m\ell}{(J + m\ell^2)M} u \cos x_1 &:= f_2(x_1, x_2, u). \end{aligned} \quad (7.81)$$

By linearising (7.81) in a neighbourhood of the equilibrium point $x_1 = x_2 = 0$ corresponding to the input $u(\cdot) = 0$, one obtains

$$\dot{\mathbf{x}} = \left[\frac{\partial f_i}{\partial x_j} \right]_{\substack{\mathbf{x}=0 \\ u=0}} \mathbf{x} + \left[\frac{\partial f_i}{\partial u} \right]_{\substack{\mathbf{x}=0 \\ u=0}} u = \begin{bmatrix} 0 & 1 \\ \frac{g}{\ell'} & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ -\frac{1}{M\ell'} \end{bmatrix} u = F\mathbf{x} + \mathbf{g}u, \quad (7.82)$$

where we assumed

$$\ell' := \frac{J + m\ell^2}{m\ell}.$$

Since F has an eigenvalue with positive real part (in fact, positive real), the origin is not a stable equilibrium point either for the linearised system or for the nonlinear one. We want to stabilise the linearised system (7.82) assuming that the only measurable output is the angular position of the pole, i.e.,

$$y = [1 \quad 0] \mathbf{x} = H\mathbf{x}. \quad (7.83)$$

Note that it is not possible to stabilise (7.82) with a simple static output feedback, since for every $k \in \mathbb{R}$ the matrix $F + \mathbf{g}kH$ takes the form

$$\begin{bmatrix} 0 & 1 \\ \alpha & 0 \end{bmatrix}$$

and hence has at least one eigenvalue with nonnegative real part.

Since the angular velocity (which is one of the state components) is not measurable, we will make use of a state-observer and then of a state-feedback law that makes use of the estimated state vector. By making use of the separation principle, we will proceed by independently designing an observer with pre-assigned error dynamics, and subsequently by allocating the eigenvalues of the closed loop system, characterised by matrix $F + \mathbf{g}K$.

i) *Observer design*

As (F, H) is observable, the eigenvalues of $F + LH$ can be arbitrarily allocated. By imposing that the error dynamics converges to zero more rapidly than $e^{-\alpha t}$, it is sufficient to choose L so that $\det(sI - F - LH) = (s + \bar{\alpha})^2$, $\bar{\alpha} > \alpha$. One obtains

$$L = \begin{bmatrix} -2\bar{\alpha} \\ -\bar{\alpha}^2 - \frac{g}{\ell'} \end{bmatrix}. \quad (7.84)$$

ii) *Feedback control law from the estimated state*

As $(F, \mathbf{g})$ is reachable, it is possible to arbitrarily assign the eigenvalues of $F + \mathbf{g}K$. To impose that the resulting system is stable, with eigenvalues λ_1 and λ_2 (with negative real part!), one can choose K so that $\det(sI - F - \mathbf{g}K) = (s - \lambda_1)(s - \lambda_2)$, thus obtaining

$$K = [M(g + \ell'\lambda_1\lambda_2) \quad -M\ell'(\lambda_1 + \lambda_2)]. \quad (7.85)$$

Remark The stabilisation procedure by means of an observer-based controller for the linearised system described in the previous example not only stabilises the linearised system, but also the non-linear one in a neighbourhood of the equilibrium point. Let us look at this in general terms. The nonlinear system resulting from the connection of

- a non-linear system (in Example 7.7.1, system (7.81) and (7.83)),
- an observer and a static feedback control acting on the estimated state, designed for the linearised system ((7.84) and (7.85) in the example),

is described by equations taking the following form

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t)) \\ \mathbf{y}(t) &= \mathbf{h}(\mathbf{x}(t)) \\ \dot{\hat{\mathbf{x}}}(t) &= (F + LH)\hat{\mathbf{x}}(t) - L\mathbf{y}(t) + G\mathbf{u}(t) \\ \mathbf{u}(t) &= K\hat{\mathbf{x}}(t) + \mathbf{v}(t), \end{aligned}$$

with $F = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right]_{\substack{\mathbf{x}=0 \\ \mathbf{u}=0}}$, $G = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right]_{\substack{\mathbf{x}=0 \\ \mathbf{u}=0}}$, $H = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right]_{\mathbf{x}=0}$, namely

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), K\hat{\mathbf{x}}(t) + \mathbf{v}(t)) \quad (7.86)$$

$$\mathbf{y}(t) = \mathbf{h}(\mathbf{x}(t)) \quad (7.87)$$

$$\dot{\hat{\mathbf{x}}}(t) = (F + LH)\hat{\mathbf{x}}(t) + GK\hat{\mathbf{x}}(t) + G\mathbf{v}(t) - L\mathbf{y}(t). \quad (7.88)$$

By linearising (7.86) and (7.87) around $\mathbf{x} = \hat{\mathbf{x}} = 0$, equilibrium point corresponding to $\mathbf{v}(\cdot) = 0$, one has

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right] \mathbf{x}(t) + \left[\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right] \left[\frac{\partial \mathbf{u}}{\partial \hat{\mathbf{x}}} \right] \hat{\mathbf{x}}(t) + \left[\frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right] \left[\frac{\partial \mathbf{u}}{\partial \mathbf{v}} \right] \mathbf{v}(t) \\ &= F\mathbf{x}(t) + GK\hat{\mathbf{x}}(t) + G\mathbf{v}(t) \end{aligned} \quad (7.89)$$

$$\mathbf{y}(t) = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right] \mathbf{x}(t) = H\mathbf{x}(t) \quad (7.90)$$

$$\dot{\hat{\mathbf{x}}}(t) = (F + LH + GK)\hat{\mathbf{x}}(t) - LH\mathbf{x}(t) + G\mathbf{v}(t). \quad (7.91)$$

Such equations coincide with those of the linear system Σ_{global} obtained by connecting the linearised system with its observer-based controller. If the global linear system is made asymptotically stable by the observer-based controller, Lyapunov linearisation theorem allows us to deduce that the origin is an asymptotically stable equilibrium point also for the non-linear system (7.86)÷(7.88), whose linearised version is Σ_{global} .

Naturally, while the domain of attraction of the origin for Σ_{global} is the entire state-space, for system (7.86)-(7.88) the origin is only locally asymptotically stable. By referring, in particular, to Example 7.7.1, this implies that the state of the overall regulated system (i.e., (7.86)-(7.88)) may not converge to the equilibrium point if the initial state of system (7.81) and/or the initial estimation error are not sufficiently close to zero.

- EXERCISE 7.7.1 Verify that the separation property remains valid even when the observer-based controller relies on a reduced-order observer.

‡ Hint: by referring, for the sake of simplicity, to the system with matrices taking the form (7.60), verify that the estimate $\hat{\mathbf{w}}$ satisfies the equation

$$\hat{\mathbf{w}}(t+1) = (A_{11} + LA_{21})\hat{\mathbf{w}}(t) - LA_{21}\mathbf{w}(t) + A_{12}\mathbf{y}(t) + B_1\mathbf{u}(t).$$

By supposing that $\mathbf{u}(t)$ is the sum of an external input $\xi(t)$ and a term which is obtained by feedback from the estimated state $\mathbf{u}(t) = K_1\hat{\mathbf{w}}(t) + K_2\mathbf{y}(t) + \xi(t)$ and by assuming as state vector of the overall system

$$\begin{bmatrix} \mathbf{w}(t) \\ \mathbf{y}(t) \\ \hat{\mathbf{w}}(t) - \mathbf{w}(t) \end{bmatrix} := \begin{bmatrix} \mathbf{w}(t) \\ \mathbf{y}(t) \\ \mathbf{e}_w(t) \end{bmatrix}$$

verify that

$$\begin{bmatrix} A_{11} + B_1K_1 & A_{12} + B_1K_2 & B_1K_1 \\ A_{21} + B_2K_1 & A_{22} + B_2K_2 & B_2K_1 \\ 0 & 0 & A_{11} + LA_{21} \end{bmatrix}$$

is the state-update matrix of the global system.

- EXERCISE 7.7.2 [OBSERVER-BASED CONTROL AND DYNAMIC OUTPUT FEEDBACK] In the system of Figure 7.7.1 modify the connections, by assuming to supply as input to the observer the term $K\hat{\mathbf{x}}(t)$. One obtains in this way the system of Figure 7.7.5.

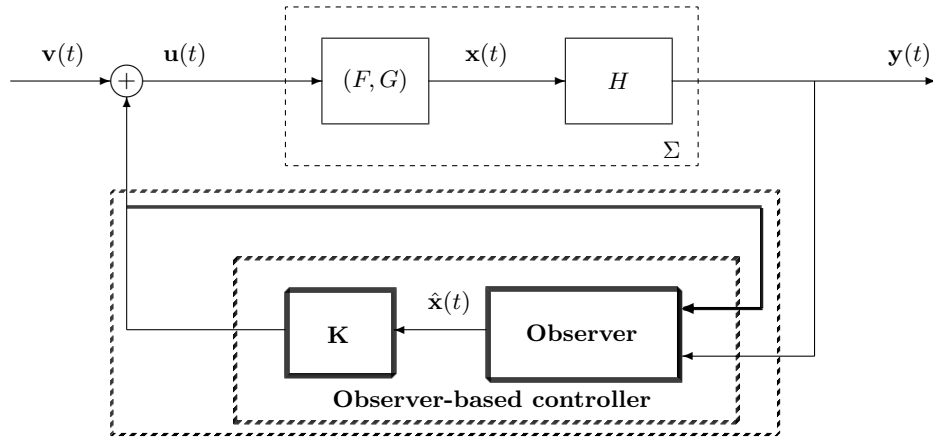


Figure 7.7.5

- Verify that the state-update matrix of the new global system is the same as the one for the global system considered in Figure 7.7.1.

- (ii) If system Σ is stabilizable and detectable, there exists a dynamic output feedback that makes the resulting system asymptotically stable.
- (iii) Do the transfer matrices of the global systems in the block schemes of Figures 7.7.1 and 7.7.5 coincide?
- EXERCISE 7.7.3 [DEAD-BEAT OBSERVER-BASED CONTROL] An observer-based controller for a discrete-time system $\Sigma = (F, G, H)$ is a dead-beat observer-based controller if it allocates all the eigenvalues of the overall system in the origin (namely if system Σ_{global} has finite memory). Verify that a necessary and sufficient condition for the existence of a dead-beat observer-based controller is that Σ is controllable and reconstructable.

7.8 References

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